

Environmental costs in second-hand clothing purchase: A voucher choice experiment

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Abstract

Although consumers may wish to act in environmentally responsible ways, informational barriers can prevent them from accurately understanding the true environmental costs of their consumption choices. We investigate how presenting different types of environmental costs (CO2 emissions, energy consumption, or water usage) affects willingness to purchase second-hand clothing. We measure changes in stated purchasing intentions and, using multiple price list, incentivized choices in a large, nationally representative sample of Italians ($N = 10,496$). We report three main findings. First, all information treatments promote second-hand adoption, but the choice of metric matters: water-based information generates the strongest effects than CO2—the dominant metric in sustainability communication. Second, pro-environmental behavior amplifies responsiveness, particularly for water-based information, while need for uniqueness acts as a barrier, with high-NFU individuals requiring more compensation to switch to second-hand. Third, local environmental conditions do not systematically moderate treatment effects. Our findings indicate that reframing environmental information from CO2 to more tangible metrics like water can meaningfully enhance sustainability campaigns, though reaching beyond already environmentally engaged consumers remains an open challenge.

Keywords: Second-hand consumption · Information provision · Environmental cost · Voucher choice · Framed field experiment

JEL Codes: D12 · D83 · D91 · Q50

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project, all authors contributed equally to the overall research project. For this specific paper, MZ secured funding and provided overall supervision; LC conducted and wrote the literature review; LL designed the experiment, analyzed the data, and drafted the remaining sections of the manuscript. All authors reviewed and approved the final version.

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1 Introduction

The fashion industry is one of the biggest markets worldwide. Two independent research reports by Euromonitor and McKinsey estimate that in 2021 it generated globally between \$1.7 trillion and \$2.5 trillion in revenues.¹ This economic turnover comes however with massive costs for the environment. Niinimäki et al. (2020) estimate that the fashion industry is responsible for 8–10% of global CO₂ emissions (4–5 billion tonnes annually), surpassing the combined emissions of international aviation and maritime shipping (Leal Filho et al., 2022), and for approximately 20% of global industrial water pollution. Water consumption is equally striking: the fashion industry uses around 93 billion cubic meters of water per year, with a single cotton t-shirt requiring up to 2,700 litres to produce (European Parliament, 2023). Beyond emissions and water use, overproduction generates roughly 92 million tonnes of textile waste annually, a figure projected to reach 134 million tonnes by 2030 (Niinimäki et al., 2020).

This environmental burden is exacerbated by the proliferation of fast fashion (Bianchi, 2018). Since 2000, clothing production has roughly doubled while the average number of times a piece of clothing is worn has declined by around 36% (Leal Filho et al., 2022). More clothes are produced, used briefly, and discarded – a dynamic that continuously generates new environmental costs. Extending the active life of clothes already in circulation, therefore, can directly reduce these costs by deferring or displacing the production of new items. The acquisition of second-hand clothing is the most direct mechanism to achieve this (Guiot and Roux, 2010). Evidence suggests

¹ <https://fashionunited.com/statistics/global-fashion-industry-statistics>

that purchasing 100 second-hand clothes is estimated to displace between 60 and 85 new clothing purchases, and that such displacement reduces toxic emissions caused by polyester cotton clothes by 45% and global warming impact of producing cotton t-shirts by 14% (Farrant et al., 2010). Recently, Klooster et al. (2024) estimated that choosing second-hand over new clothing reduces the impact of fashion on climate change by up to 42% and water scarcity by 35–53%.

Despite the beneficial environmental impact of second-hand clothing, adoption rates worldwide remain generally low. In the European Union, only 22% of post-consumer textile waste is collected separately for recycling or reuse, while the rest is often burned or deposited in landfills (European Commission Press Release, 2023). Understanding what drives demand for second-hand clothing, and what inhibits it, is therefore a precondition for designing policies capable of accelerating the transition away from new clothing consumption. In this respect, a determinant that is receiving increasing attention is consumers’ awareness of the environmental harms caused by the fashion industry. Empirical evidence suggests that consumers that are aware of the environmental costs associated with the production of new clothes show higher intention to purchase them second-hand (Borusiak et al., 2020; Pretner et al., 2021; Negash and Akhbar, 2024). However, research also shows that such awareness among consumers is currently low (Goworek et al., 2012; Harris et al., 2016) and that it may be difficult to increase it. For example, several contributions have tested the effect of providing information about carbon footprint, and they have found that consumers report difficulties in understanding CO₂-based information (Guenther et al., 2012; Hartikainen et al., 2014; Dreist et al., 2024). Consequently, a stream of the literature has started investigating how such information can be expressed in metrics that can be more easily digested by consumers, for instance equating it with miles driven per gallon (Larrick et al., 2015) or minutes of light-bulb usage (Camilleri et al., 2019).

This paper investigates how the willingness to purchase second-hand clothing is affected by the expression of environmental costs through a framed field experiment involving two tasks.

In the first task, we ask respondents to guess the environmental cost avoided by not producing a new piece of clothing, in a quiz manner, and then we inform them about the correct figure. Depending on treatment, the cost is expressed as kilograms of CO₂ emitted (Kg CO₂e), kilowatt-hour of electric energy (kWh), or liters of water (ℓ). All these externalities are involved in the production of new clothing and, unlike CO₂, electric energy and water are everyday resources believed to be more relatable to individuals. Moreover, to facilitate comprehension of the measurement units involved, each is supplemented with an illustrative equivalence in a metric consistent with the environmental cost under consideration (e.g., 1 kg CO₂e is equated to the distance in km that a medium-sized car has to drive to generate it). We also ask respondents whether, based on the information received, they would be less, equally, or more likely to purchase second-hand in their daily lives. Respondents in the control skip this task.

The second task involves a voucher choice. Respondents are presented with a multiple price list involving pairs of vouchers of different amounts, where one can be redeemed in a traditional clothing retailer and the other in a second-hand shop. For half of the sample, we kept the traditional-store voucher fixed and varied the amounts of the second-hand one, thereby eliciting the willingness to accept (WTA) to switch consumption from first-hand to second-hand clothing. For the other half, instead, we did the opposite and thus elicited willingness to pay (WTP) to switch from second-hand to first-hand clothing.

We find that, consistent with our hypotheses, providing information in terms of water or energy usage generates stronger effects on stated intention to purchase second-hand clothing and voucher choices than CO₂-based information. Water-based information yields the largest treatment effects, with 43% of respondents declaring to be more likely to purchase (compared to 35% for energy and 30% for CO₂), and reducing willingness to accept to forgo second-hand consumption by approximately €4 relative to the control group. Energy information shows intermediate effectiveness, while CO₂ emerges as the weakest treatment despite being the most commonly used metric in sustainability communication. Interestingly, respondents exhibit consistent voucher choices across the two versions of the task, that is, relative to whether the first-hand voucher is set as the fixed option. This suggests that our results are not driven by a framing or an endowment effect, even though the magnitude of WTA and WTP varies with which option is designated as the option.

We also document important heterogeneity in treatment effects. Information provision is most effective among individuals who initially underestimate environmental benefits of favoring second-hand over first-hand (approximately 62% of our sample), while those with accurate prior beliefs show substantial resistance to updating their stated intention. Treatment effects are highest among people scoring high on the pro-environmental behaviors scale (PEBS; [Menardo et al., 2020](#)) and hindered by those exhibiting a high need for uniqueness (NFU scale; [Ruvio et al., 2008](#)). Finally, considering that local environmental conditions may influence how respondents respond to our treatments, we have linked their choices to administrative data on water scarcity, energy poverty, and air quality, at the provincial or regional level. Contrary to our expectations, we find limited evidence that local environmental conditions moderate treatment effects. Similarly, store proximity influences baseline voucher preferences but does not moderate information provision effects, indicating that our treatments are effective regardless of the perceived accessibility of the conduct.

The paper proceeds as follows. [Section 2](#) reviews the relevant literature on environmental information provision. [Section 3](#) details our experimental design. [Section 4](#) presents our main findings on stated intentions and voucher choices. [Section 5](#) examines heterogeneous treatment effects by prior beliefs and individual characteristics, and explores potential moderators including geographic location, store proximity, and local environmental conditions. [Section 6](#) discusses the policy implications of our findings as well as the study limitations and some directions for future research.

2 Literature Review

A well-established literature has investigated the effectiveness of CO₂-based information provision on consumer choices (see, e.g., the literature reviews by [Liu et al., 2016](#); [Rondoni and Grasso, 2021](#)). What these contributions generally find is that information referencing CO₂ emissions can be effective in fostering sustainable consumption, but its effectiveness appears modest. To our knowledge, the study by [Pretner et al. \(2021\)](#) is one of the few to examine the effect of CO₂-based information on second-hand clothing choices. The study finds informing participants about the number of emissions prevented by choosing a second-hand hoodie over a new one increased willingness to pay (WTP) from \$11.90 to \$15.30 in the US sample and from €18.00 to €19.60 in the Italian sample. This limited effectiveness may be due to the difficulty that individuals report

in relating to CO₂-based information (Guenther et al., 2012; Hartikainen et al., 2014; Rondoni and Grasso, 2021; Dreist et al., 2024).

Previous research has found that communicating metrics in a comprehensible and relatable way enhances information provision (Camilleri and Larrick, 2014; Larrick et al., 2015; Ungemach et al., 2018). In effect, carbon labels are more effective in fostering sustainable choices when they complement CO₂ information with a traffic light system (Thøgersen and Nielsen, 2016; Lohmann et al., 2022) or provide an equivalence with other, more relatable measures, such as monetary saving (Newell and Siikamäki, 2014) or minutes of usage of a light bulb (Camilleri et al., 2019). Notably, Camilleri et al. (2019) find that a label reporting the light-bulb minutes equivalent of a serving of food items (as well as its ranking in terms of carbon footprint) led participants to purchase fewer cans of beef soup and report more accurate estimates for CO₂ emissions caused by beef.

Building on these findings, in our experiment, we provide information on the environmental damage that can be avoided by purchasing a second-hand piece of clothing rather than a new one by expressing the damage in alternative environmental externalities (CO₂, energy, or water) and converting the metric for each externality into a familiar, but closely related, equivalent metric. The choice of energy and water as alternatives to CO₂ is motivated by the observation that, while CO₂ emissions are frequently used as the primary or sole indicator of environmental impact in sustainability communication, clothing production generates multiple distinct environmental externalities, notably energy consumption and water usage.

To enhance information clarity, we express the environmental cost in terms of habitual or familiar actions, making the damage more concrete. We restated kilograms of CO₂ as those emitted by an equivalent number of kilometers driven by a medium-sized gasoline car (Pace et al., 2025 for a similar conversion rationale). Regarding energy and water saving, we adopt respectively the equivalence of kWh to hours of a 100-watt light-bulb usage, and liters to minutes of open kitchen tap, since both are actions that individuals can easily relate to and experience in their daily lives (details of the implementation are discussed in Section 3.2).

By varying the type of environmental cost information, the present experiment contributes to the discussion on how to design informational campaigns aimed at raising environmental awareness to encourage consumers to adopt greener consumption patterns. Additionally, the experiment sees two elements of novelty. First, rather than simply providing information about environmental costs, it administers an information provision task in the form of a quiz, intending to enhance the respondent’s curiosity, engagement, and deliberation. Second, the experiment addresses the topic of conscious consumption choices, particularly in the textile industry, by adopting a voucher choice task. We are aware of other papers relying on voucher choices, such as in meat consumption (Pizzo et al., 2024), yet none do it for clothing.

3 Experimental Design

The information provision and voucher choice tasks were embedded in a larger survey about second-hand. The study received a Certificate of Good Standing by the Board for Ethical Questions in

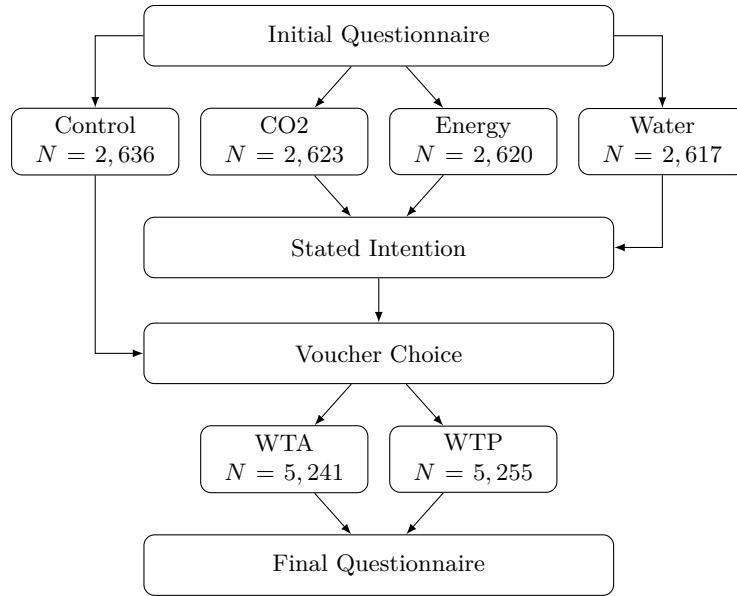
Science of the University of Innsbruck and was pre-registered on Open Science Framework (OSF).² The full text of the information provision and voucher choice tasks is available, respectively, in [Section A.1](#) and [Section A.2](#), and, together with the text of the full survey, in the project OSF repository.

3.1 Procedure

The survey was implemented and administered by the survey company Scenari from October to December 2024 to members of their panel of respondents, using the computer-assisted web interviewing (CAWI) methodology. Respondents were recruited using a stratified sampling approach to ensure that the sample ($N = 10,496$) was representative at the provincial level with respect to the following sociodemographic variables: age ($m = 49.67$, $sd = 15.62$), gender (percentage of men: 48.83%; percentage of women: 51.17%), education (percentage of people holding a university degree or above: 36.81%) and geographic area (Northwest: 26.93%; Northeast: 19.76%; Center: 19.63%; South: 22.77%; Islands: 10.91%); see [Table B.1](#) in [Appendix B](#) for the full descriptive statistics.

During the survey, respondents were unaware of the upcoming questions and could not go back to amend their answers. As a reward for completion, respondents were paid €4. Additionally, participants could win a voucher (€0–20) based on their choices in the voucher task and a random draw (see details in [Section 3.3](#)). [Figure 1](#) provides a visual summary of the experimental design, illustrating the sequence of tasks from random assignment to treatment through the elicitation of stated intentions and voucher choices.

Figure 1: Experimental Design Flowchart



² The OSF repository of the project can be accessed here: osf.io/wzcqh

3.2 Information provision task

The information provision task involved guessing how much resources could be saved in terms of CO₂, electricity, or water when switching to second-hand clothing (from first-hand clothing). To aid respondents in this task, we supplemented the information with equivalences using measures that we felt could be more familiar and relatable to respondents. Specifically, to put kg CO₂e in perspective, we told respondents that 1 kg CO₂e equates to 6 kilometers driven by a medium-sized gasoline car. Figure 2 shows an illustration of the task for the CO₂e treatment, translated from Italian (see Section A.1 for the versions proposed in the other two treatments).

Figure 2: Illustration of the information provision task

(a) Quiz

Before answering the next question, we want to inform you of the value of one kilogram of carbon dioxide emitted (CO₂e).

1 kg CO₂e = 6 km driven by a medium-sized gasoline car

When you buy a second-hand piece of clothing, do you know how much CO₂ emissions you can avoid by not producing a new one? About: ____ kg CO₂e

(b) Information provision and intention elicitation

The exact estimate is about 4 kg CO₂e, or about 24 km traveled by a medium-sized gasoline car. Based on this information, are you more or less inclined to buy second-hand in your daily life?

- ☐ I am more inclined
- ☐ I am equally inclined
- ☐ I am less inclined

In the energy treatment, respondents were informed that 1 kWh equates to 10 hours of use of a 100-watt bulb. In the water treatment, they learned that 10 liters equate to 2 minutes of an open kitchen tap. Once they had expressed their guess, respondents were shown the true values in a separate screen: 4 kg CO₂e, 17 kWh, and 340 liters, respectively. These values and equivalences adopted in the task were based on estimates from various scientific and institutional sources.³ Finally, following the information provision, respondents were asked to state whether they were less, equally, or more inclined to buy second-hand clothing themselves.⁴

³ We based the equivalence 1 kg CO₂e = 6 km traveled by an average car on that provided by a 2023 UK governmental report (<https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2023>). This estimate is roughly consistent with figures provided by other sources, such as CO₂everything (around 5 km; <https://www.co2everything.com/co2e-of/cotton>). For the equivalences related to energy and water, we referred to those provided by WRECC.com (<https://www.wrecc.com/what-uses-watts-in-your-home/>) and the UK Government (<https://assets.publishing.service.gov.uk/media/5a75bd8940f0b67b3d5c8da0/gbs-taps-automatic-sprays-showers-urinal-2015.pdf>), respectively. For the amount of resources (CO₂, electricity, and water) saved when adopting second-hand clothing, we referred to estimates indicated in a report by ThredUp (https://cf-assets-tup.thredup.com/resale_report/2023/thredUP_2023_Resale_Report_FINAL.pdf).

⁴ We did not elicit the respondent's posterior or updated beliefs. This is because the updating boiled down to a simple figure, and we conjectured that the respondents would have been able to input the exact true figure because they had just seen it. Besides, a potential downside of measuring both prior and posterior beliefs is that the latter may be biased by consistency or experimenter demand effects (Haaland et al., 2023).

3.3 Voucher choice task

After the information provision task, respondents are presented with a multiple price list (MPL) in which they can express their preferences among pairs of vouchers of different amounts to be redeemed in either a first-hand or a second-hand store. We used two different versions of this MPL (see [Section A.2](#) for the experimental screens). In the first version ([Figure 3](#)), the value of the first-hand voucher (Voucher A) is fixed across rows at €10, and the task elicits the respondent’s willingness to accept (hereinafter WTA) compensation to switch from a first-hand voucher to a second-hand one (in this case, Voucher B) valued at 0 to €20. In the second version, with Voucher A being the second-hand voucher, the task elicits the respondent’s willingness to pay (hereinafter WTP) to upgrade from a second-hand voucher to a first-hand voucher, keeping the voucher values identical. The conceptual distinction between these two measures stems from the nature of the fixed option: WTA captures the compensation required when moving from a guaranteed option typically regarded as superior (first-hand) to an alternative regarded as inferior (second-hand), while WTP captures the premium one is willing to pay for the reverse transition.⁵ The first version of the task is thus the most intuitive for the interpretation of results, as it sets the first-hand store as the respondents’ status quo; it is also the most widespread in similar experiments (for example, [Allcott and Taubinsky, 2015](#)). We chose to administer also the version with second-hand as status quo to provide a more complete estimation of the treatment effects and potentially provide methodological insights on this type of policy evaluation (e.g., consistency in treatment effects across the two MPLs or lack thereof, and the relative magnitude of WTA versus WTP responses).

Respondents were required to select a single switching point in the price list, transitioning directly from one voucher option to the other, with no possibility of multiple or inconsistent switches. The voucher could be redeemed at two prominent first-hand and second-hand stores. We carefully selected these two stores to make sure they were comparable. Both stores have similar price levels and are spread across Italy, so they are widely available to our respondents. This was important because our participants come from all over the country. We will discuss how store proximity and accessibility might affect our results in [Section 5](#).

Figure 3: MPL for the voucher task

Voucher A	Voucher B	Your choice	
€10	€0	☐ Voucher A	☐ Voucher B
€10	€5	☐ Voucher A	☐ Voucher B
€10	€10	☐ Voucher A	☐ Voucher B
€10	€15	☐ Voucher A	☐ Voucher B
€10	€20	☐ Voucher A	☐ Voucher B

On a different screen, participants could opt out of the lottery. We included this option to see if information about environmental costs would encourage people to reduce their consumption rather than favoring second-hand items.

⁵ For a more in-depth discussion of the differences between WTA and WTP, see [Bateman et al. \(2002\)](#).

3.4 Registered hypotheses

Our paper aims to answer the following research questions: *i)* Does information provision on environmental costs foster intention to buy second-hand goods over new ones? *ii)* Does information provision on environmental costs foster willingness to accept for second-hand over first-hand, and willingness to pay for first-hand over second-hand? *iii)* Do the different environmental cost framing have a different impact on stated intention and WTA/WTP as compared to the others?

Following our research questions, we first hypothesize that, regardless of the externality under consideration (CO₂, energy, water), the impact of information provision on the change in the stated relative intention to purchase second-hand clothing is higher the higher the difference between the respondent's initial belief and the revealed true value. Specifically, we conjecture that, when respondents realize that they have underestimated the environmental cost of first-hand clothing, they might, in response, be willing to change their consumption pattern toward second-hand; likewise, those who realize they overestimated the cost might persevere in or increase their consumption of first-hand clothing. Accordingly, our hypothesis is:

H.1 The further from the true value is the respondent's belief...

H.1.a in positive (overestimation of the true value), the lower the intention to purchase second-hand clothing.

H.1.b in negative (underestimation of the true value), the higher the intention to purchase second-hand clothing.

Secondly, we hypothesize that the effectiveness of information provision on the intention to buy second-hand clothing is higher when it refers to water rather than energy and to energy rather than CO₂ emissions, since the three environmental externalities differ in the degree to which the respondent can "relate" to them.

H.2 The frequency of respondents declaring a higher intention to purchase second-hand goods is...

H.2.a higher in Water than Energy treatment

H.2.b higher in Energy than CO₂ treatment

H.2.c higher in Water than CO₂ treatment

We also hypothesize that WTA (WTP) differentials between first-hand (second-hand) and second-hand (first-hand) clothing will be sensitive to the type of environmental externality and the distance between the respondent's guess in the quiz and the true value revealed with the information provision. Specifically, based on our hypotheses *H.1* and *H.2*, we hypothesize the following:

H.3 We expect the WTA differential (with first-hand as the fixed option) to be lower (respondents accept a lower value of the voucher to switch to second-hand)...

H.3.a the more reliable is the environmental cost.

H.3.b the higher the underestimation of the true value.

H.4 We expect the WTP differential (with second-hand as the fixed option) to be higher (respondents accept a higher value of the voucher to switch to first-hand)...

H.4.a the more relatable is the environmental cost.

H.4.b the higher the underestimation of the true value.

4 Main Results

4.1 Stated Intentions

Participants in the three treatments were asked their intention to buy second-hand after the information provision, we find that a substantial proportion of respondents (53.84%) declared that they would be equally likely as before receiving the information to adopt second-hand clothing in the future, followed by a proportion of participants stating they were more likely to do so (36.34%). Only a minority (9.82%) stated a lower propensity after the belief updating.

To test Hypothesis *H.1*, we coded respondents' initial beliefs (guesses) as correct, overestimated, or underestimated by creating a tolerance interval of $\pm 15\%$ around the true value. This classification allowed us to identify which respondents had overestimated and which had underestimated the true environmental benefit of purchasing second-hand clothing. We then employed chi-squared tests to examine whether the distribution of stated intentions (i.e., equally likely, more likely, or less likely to adopt second-hand) differed significantly between those who overestimated versus those who did not, and between those who underestimated versus those who did not. The results of these tests are reported in [Table 1](#); furthermore, [Figure B.1](#) in [Appendix B](#) shows the same trend.

Table 1: Intention Distribution by Belief Accuracy

Intention	Belief Accuracy		
	Underestimate	Accurate	Overestimate
Less likely	551 (11.26%)	62 (6.52%)	159 (7.89%)
Equally likely	2,441 (49.88%)	624 (65.62%)	1,167 (57.92%)
More likely	1,902 (38.86%)	265 (27.87%)	689 (34.19%)
Total	4,894 (100%)	951 (100%)	2,015 (100%)

Note: Cross-tabulation of stated intention changes by belief accuracy category. Belief accuracy determined using $\pm 15\%$ tolerance interval around true values. Percentages represent column proportions within each belief accuracy category. Pearson $\chi^2 = 103.41$, $p < 0.001$.

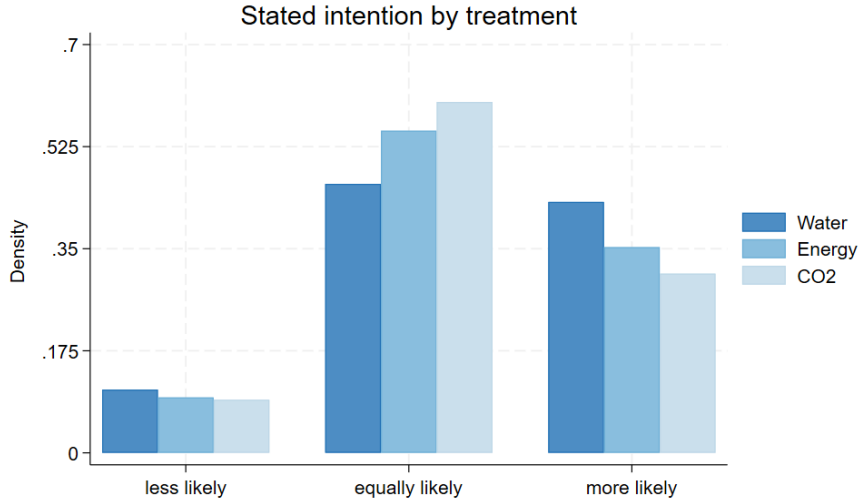
A chi-squared test shows that the intention and guesses are significantly associated, that is, that the relative stated intention to adopt second-hand changes based on underestimation or overestimation of costs (Pearson $\chi^2 = 103.41$, $p < 0.001$). Information provision appears most effective among those who initially underestimated the environmental benefit: 38.86% of underestimators reported being more inclined to adopt second-hand clothing, compared to 34.19% of overestimators. Conversely, respondents whose initial beliefs were accurate proved to be the most resistant to changing their stated intentions, with 65.62% reporting no change in their propensity to purchase second-hand. Notably, relatively few respondents across all groups became less inclined to

adopt second-hand following the information provision (ranging from 6% to 11%), suggesting that providing environmental information rarely produces counterproductive effects. From a policy perspective, these results highlight that underestimators represent both the largest segment of our sample (62.26%) and the group most responsive to information provision, suggesting considerable potential for interventions targeting this population.

Result 1 (H.1): *Information provision has heterogeneous effects depending on prior beliefs, with underestimators showing significantly greater responsiveness than overestimators or those with accurate initial beliefs.*

Moving on to testing Hypothesis *H.2*, we examine whether the type of environmental externality presented influences respondents' stated intentions to adopt second-hand clothing. Our hypothesis predicted that information framed in terms of water and energy would be more effective than CO₂-based information in increasing adoption intentions. As illustrated in Figure 4, the patterns just identified exhibit meaningful variation across the three treatment conditions, consistent with our directional prediction. In the CO₂ treatment, 60.16% of respondents reported being equally likely, 30.73% more likely, and 9.11% less likely to adopt second-hand clothing following the information provision. In the energy treatment, the corresponding figures were 55.23%, 35.27%, and 9.50%, respectively. The water treatment yielded the most favorable distribution, with 46.12% remaining equally likely, 43.03% becoming more likely, and 10.85% becoming less likely to purchase second-hand items.

Figure 4: Stated intention to adopt second-hand clothing across treatments



Note: Histogram showing the distribution of stated intention changes (less likely, equally likely, more likely to adopt second-hand) across treatment conditions (Water, Energy, CO₂). Density represents the proportion of respondents in each category within each treatment group.

A chi-squared test confirms that these differences across treatments are statistically significant (Pearson $\chi^2 = 109.56$, $p < 0.001$; see Table B.2 in the Appendix B for detailed results). The distribution reveals a clear gradient in effectiveness, with water-related information generating the most positive response, followed by energy, and then CO₂, precisely as hypothesized. While the

proportion of respondents becoming less inclined remains fairly stable across treatments (around 9-11%), the key difference lies in the shift from “equally likely” to “more likely” categories, with water information proving most effective in moving respondents toward increased adoption intentions.

Result 2 (H.2): *The type of environmental externality communicated significantly affects stated behavioral intentions. Water-based information is most effective in increasing adoption intentions (43.03% more likely), followed by energy (35.27%), and CO2 (30.73%), supporting the hypothesis that more tangible and relatable environmental metrics generate stronger behavioral responses.*

To corroborate our initial results, we estimate a series of ordered logistic regressions examining how stated intention to purchase second-hand clothing varies with both the type of environmental information provided and the magnitude of belief updating triggered. [Table 2](#) reports the main specifications of these regressions, while the complete models including sociodemographic controls are presented in [Table B.7](#) in [Appendix B](#).

Table 2: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	(1)	(2)	(3)
Delta Belief	-0.004** (0.002)	-0.001 (0.002)	-0.006*** (0.002)
Water		0.371*** (0.056)	0.367*** (0.059)
Energy		0.135*** (0.052)	0.132** (0.055)
Cut 1	-2.226*** (0.038)	-2.065*** (0.046)	-2.327*** (0.142)
Cut 2	0.553*** (0.024)	0.726*** (0.039)	0.953*** (0.139)
Controls	No	No	Yes
Observations	7860	7860	7860

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely). *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, and macro-region of residence (baseline categories set to the mode; full specification in [Table B.7](#)).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Model (1) examines the role of belief accuracy across all treatments. The negative and significant coefficient on Delta Belief (-0.004 , $p < 0.005$) indicates that stated intention to increase second-hand purchasing decreases as the distance between the respondent’s prior belief and the true value increases, regardless of which environmental resource was targeted. This result provides additional support for Result 1, confirming that respondents with more accurate initial beliefs are less responsive to information provision.

Model (2) incorporates treatment indicators to test whether the type of environmental externality affects stated intentions, using CO2 as the baseline category. Consistent with Result 2, both

Water and Energy treatments generate significantly more positive responses than the CO2 treatment. The Water treatment exhibits the strongest effect (0.371, $p < 0.01$), followed by the Energy treatment (0.135, $p < 0.01$). These findings reinforce our earlier descriptive evidence that more tangible and relatable environmental metrics are more effective in shifting behavioral intentions.

Model (3) adds the full set of sociodemographic controls (age, gender, income, education, employment status, pro-environmental behavior, need for uniqueness, past adoption of second-hand, and macro-region of residence; see [Table B.7](#) for the complete specification). The treatment effects remain virtually unchanged: Water (0.367, $p < 0.01$) and Energy (0.132, $p < 0.05$) continue to outperform CO2. Delta Belief becomes more strongly significant (-0.006 , $p < 0.01$), suggesting that once observable heterogeneity is accounted for, belief accuracy plays a clearer role in moderating responsiveness to information provision.

4.2 Willingness to Accept and Willingness to Pay Analysis

We now turn to testing Hypotheses *H.3* and *H.4*, which concern how respondents' willingness to accept (WTA) compensation to switch from first-hand to second-hand vouchers, and their willingness to pay (WTP) a premium to switch from second-hand to first-hand vouchers, are affected by the type of environmental information provided and the magnitude of belief updating required. Recall that WTA is elicited using a multiple price list where first-hand is the fixed option (Voucher A), while WTP is elicited with second-hand as the status quo. Both measures are computed as the midpoint of the interval in which the respondent switches from one option to the other, normalized as the difference from the fixed option value of €10, resulting in variables ranging from -€12.5 to €12.5.

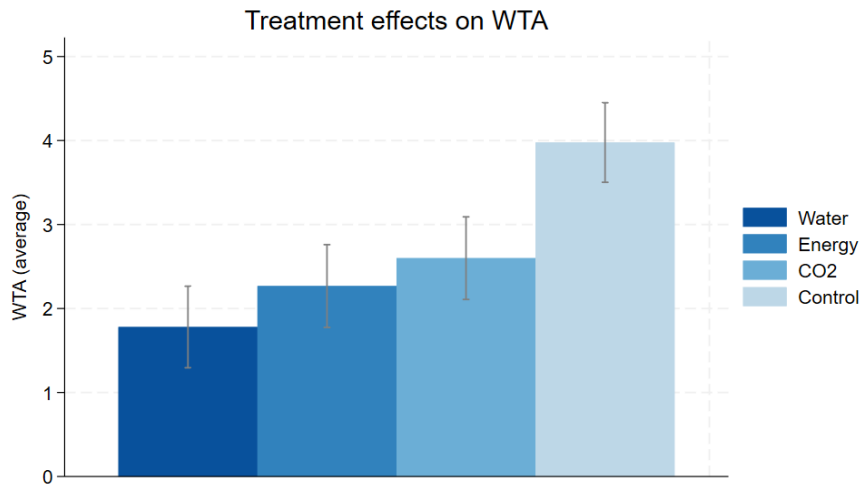
[Figure 8a](#) and [Figure 8b](#) illustrate how WTA and WTP vary across treatments. We begin by examining simple mean comparisons before moving to multivariate regression analysis to disentangle the effects of treatment assignment and belief accuracy.

4.2.1 Willingness to Accept: The Effect of Information Provision

We first examine whether providing environmental information reduces respondents' WTA to switch from first-hand to second-hand vouchers. [Table 3a](#) presents Bonferroni-adjusted pairwise mean comparisons across treatment groups. All three information treatments (Water, Energy, and CO2) significantly reduce WTA compared to the Control group, with the Water treatment showing the largest effect ($-\text{€}2.20$), followed by Energy ($-\text{€}1.71$) and CO2 ($-\text{€}1.38$). Notably, the differences between the three information treatments are not statistically significant, suggesting that while information provision itself is effective, the relative advantage of more relatable metrics is modest when examining WTA in isolation.

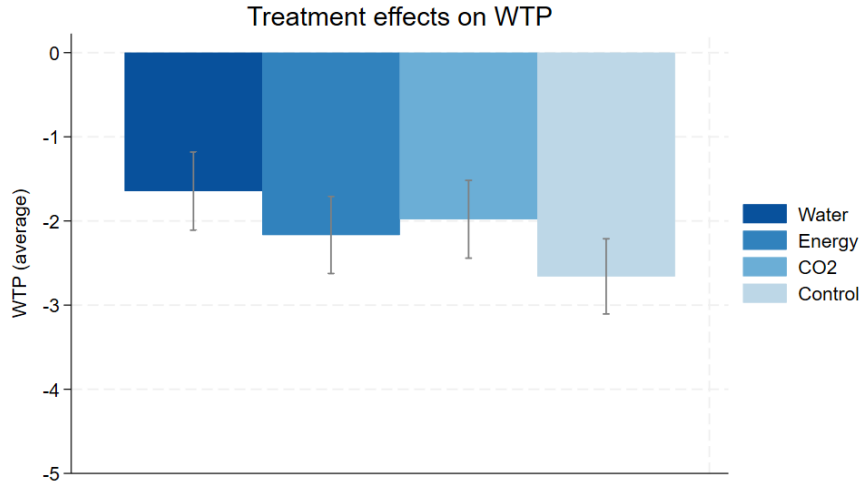
[Table 3b](#) examines whether WTA varies by initial belief accuracy. Contrary to what we observed for stated intentions, belief accuracy shows no significant impact on WTA: the differences between underestimators, accurate believers, and overestimators are all statistically indistinguishable. This pattern suggests that while belief accuracy matters for stated behavioral intentions, the more consequential voucher choice task is primarily driven by whether individuals update their intentions at all, regardless of their prior beliefs.

Figure 5: Willingness to Accept (WTA) for Second-Hand Vouchers by Treatment



Note: Bar graph showing mean WTA (willingness to accept compensation to switch from first-hand to second-hand vouchers) across treatment conditions. WTA computed as the midpoint of the switching interval in the multiple price list, normalized to baseline €10. Negative values indicate lower WTA (greater willingness to accept second-hand). Error bars represent 95% confidence intervals.

Figure 6: Willingness to Pay (WTP) for First-Hand Vouchers by Treatment



Note: Bar graph showing mean WTP (willingness to pay premium to switch from second-hand to first-hand vouchers) across treatment conditions. WTP computed as the midpoint of the switching interval in the multiple price list, normalized to baseline €10. Positive values indicate higher WTP (greater willingness to pay for first-hand). Error bars represent 95% confidence intervals.

To further investigate the relative importance of information type versus belief accuracy, we estimate a series of Tobit regressions whose main specifications are reported in [Table 4](#) (the com-

Table 3: WTA: Pairwise Mean Comparisons (Bonferroni Adjusted)

(a) By Treatment

	Control	Water	Energy	CO2
Control	—	−2.20 ***	−1.71 ***	−1.38 ***
Water		—	0.49	0.82
Energy			—	0.33
CO2				—

(b) By Belief Accuracy

	Underestimate	Accurate	Overestimate
Underestimate	—	−0.35	−0.47
Accurate		—	−0.12
Overestimate			—

Note: Values represent the difference in mean WTA between the column category and the row category (column mean minus row mean). Belief accuracy determined using $\pm 15\%$ tolerance interval around true values. Significance levels (Bonferroni adjusted): * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

plete models including sociodemographic controls are presented in [Table B.8](#) in [Appendix B](#). Column (1) confirms that all three information treatments significantly reduce WTA relative to Control, with Water showing the strongest effect (-4.294 , $p < 0.01$), followed by Energy (-3.239 , $p < 0.01$) and CO2 (-2.531 , $p < 0.01$). Column (2) adds sociodemographic controls; the pattern remains consistent, with all three treatments retaining significance relative to Control, though the magnitudes are somewhat attenuated. Column (3) uses CO2 as the baseline and introduces Delta Belief, the standardized distance between the respondent’s prior belief and the true value. While Water remains significantly more effective than CO2 (-2.079 , $p < 0.01$), the effect of Energy is no longer statistically distinguishable from CO2. Delta Belief exhibits a negative and significant effect (-0.047 , $p < 0.05$), indicating that respondents who were further from the true value show somewhat lower WTA. Column (4) adds controls to this specification: Water remains significant (-1.572 , $p < 0.05$), but the Delta Belief coefficient becomes insignificant, suggesting that the belief accuracy effect observed in column (3) is partly absorbed by observable individual characteristics.

These findings provide partial support for our hypotheses:

Result 3 (H.3.a): *Information provision significantly reduces WTA to switch from first-hand to second-hand vouchers across all treatment conditions. Water-based information demonstrates the strongest effect, though the advantage over other environmental metrics is more modest than observed for stated intentions, with only Water showing a statistically significant difference from CO2 when controlling for belief updating.*

Result 4 (H.3.b): *Belief accuracy does not significantly affect WTA. Pairwise comparisons reveal no differences across accuracy groups, and the Delta Belief coefficient, while significant*

without controls, vanishes once sociodemographic characteristics are included.

Table 4: Tobit Regression Estimations of WTA for Second-Hand

	(1)	(2)	(3)	(4)
Water	-4.294*** (0.669)	-3.688*** (0.623)	-2.079*** (0.683)	-1.572** (0.628)
Energy	-3.239*** (0.681)	-2.145*** (0.633)	-0.963 (0.688)	0.007 (0.635)
CO2	-2.531*** (0.680)	-2.249*** (0.631)		
Delta Belief			-0.047** (0.024)	-0.015 (0.022)
Constant	7.635*** (0.492)	14.498*** (1.423)	5.373*** (0.507)	13.069*** (1.599)
Controls	No	Yes	No	Yes
Observations	5241	5074	3930	3788

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTA*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Negative values indicate lower WTA (greater willingness to accept second-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control* in columns (1)–(2); *CO2* in columns (3)–(4). *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value. *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, macro-region of residence, distance to the nearest first-hand and second-hand store, and whether the closest store is second-hand (baseline categories set to the mode; full specification in [Table B.8](#)).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Willingness to Pay: The Role of Status Quo Framing

We now examine WTP, where second-hand serves as the status quo and respondents indicate how much they would pay to switch to a first-hand voucher. It is important to note that by setting second-hand as the default in this task, we are effectively forcing a status quo that may not align with respondents’ natural reference point, potentially attenuating treatment effects relative to what we observed for WTA.

[Table 5a](#) presents Bonferroni-adjusted mean comparisons. Only the Water treatment shows a significant increase in WTP relative to Control (difference of $\text{€}1.01$), while Energy and CO2 do not differ significantly from Control. This pattern already suggests more muted effects compared to the WTA analysis.

[Table 5b](#) shows an interesting pattern: respondents with accurate initial beliefs exhibit significantly higher WTP than underestimators (difference of $\text{€}1.34$) and overestimators (difference of $\text{€}0.89$). This suggests that when second-hand is set as the default, those who already had accurate beliefs about environmental benefits are most willing to pay to revert to first-hand, possibly because they feel less need to be “nudged” away from their preferred choice.

The Tobit regressions in [Table 6](#) provide further insights into these effects ([Table 6](#) reports the

Table 5: WTP: Pairwise Mean Comparisons (Bonferroni Adjusted)

(a) By Treatment				
	Control	Water	Energy	CO2
Control	—	1.01 **	0.49	0.68
Water		—	-0.52	-0.33
Energy			—	0.19
CO2				—

(b) By Belief Accuracy			
	Underestimate	Accurate	Overestimate
Underestimate	—	1.34 ***	0.89 **
Accurate		—	-0.44
Overestimate			—

Note: Values represent the difference in mean WTP between the column category and the row category (column mean minus row mean). Belief accuracy determined using $\pm 15\%$ tolerance interval around true values. Significance levels (Bonferroni adjusted): * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

main specifications of these regressions, while the complete models including sociodemographic controls are presented in Table B.8 in Appendix B). Column (1) shows that both Water (1.574, $p < 0.001$) and CO2 (1.097, $p < 0.05$) increase WTP relative to Control, but Energy does not. Column (2) adds sociodemographic controls; the pattern is confirmed, with Water (1.293, $p < 0.01$) and CO2 (1.196, $p < 0.05$) remaining significant. Column (3) uses CO2 as baseline and introduces Delta Belief, which exhibits a positive and marginally significant effect (0.032, $p < 0.10$): respondents who were further from the true value show higher WTP to switch back to first-hand. This contrasts sharply with the WTA results and suggests that when second-hand is imposed as the status quo, those who had underestimated environmental benefits (and thus receive larger corrections) are inclined to pay more to return to first-hand options. Neither Water nor Energy significantly differs from CO2 in this specification. Column (4) adds controls: treatment effects remain insignificant relative to CO2, and the Delta Belief coefficient also becomes insignificant, mirroring the pattern observed for WTA.

These findings provide limited support for our hypotheses:

Result 5 (H.4.a): *Information provision has more limited effects on WTP than on WTA. Water and CO2 coefficients are significant, while only Water-based information shows a robust effect relative to Control in pairwise comparisons. When using CO2 as baseline, no information type demonstrates a significant advantage.*

Result 6 (H.4.b): *Belief accuracy shows an association with WTP. While pairwise comparisons reveal significant differences across accuracy groups, the Delta Belief coefficient is only marginally significant without controls and vanishes once sociodemographic characteristics are in-*

cluded.

Table 6: Tobit Regression Estimations of WTP for First-Hand

	(1)	(2)	(3)	(4)
Water	1.574*** (0.532)	1.293*** (0.495)	0.710 (0.556)	0.214 (0.517)
Energy	0.751 (0.529)	0.779 (0.494)	-0.166 (0.548)	-0.322 (0.509)
CO2	1.097** (0.532)	1.196** (0.494)		
Delta Belief			0.032* (0.017)	0.022 (0.016)
Constant	-4.206*** (0.373)	-7.740*** (1.092)	-3.328*** (0.401)	-6.578*** (1.247)
Controls	No	Yes	No	Yes
Observations	5255	5105	3930	3817

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTP*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Positive values indicate higher WTP (greater willingness to pay for first-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control* in columns (1)–(2); *CO2* in columns (3)–(4). *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value. *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, macro-region of residence, distance to the nearest first-hand and second-hand store, and whether the closest store is second-hand (baseline categories set to the mode; full specification in Table B.8).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Underlying Mechanisms: Pro-Environmental Behavior and Need for Uniqueness

To investigate more thoroughly the psychological drivers of responses to environmental information, we examine two pre-registered individual-level mechanisms: pro-environmental attitudes and need for uniqueness. We elicit those constructs using validated scales, respectively the Pro-Environmental Behavior Scale (PEBS) by Menardo et al. (2020), as adapted from Markle (2013), and the Need for Uniqueness (NFU) scale by Ruvio et al. (2008), as adapted from Tian et al. (2001). The Pro-Environmental behavior scale (hereinafter, PEB) captures respondents' general orientation toward sustainable consumption practices, which we expect to amplify responsiveness to environmental messaging. Need for Uniqueness (hereinafter, NFU), by contrast, reflects individuals' desire to differentiate themselves from others through distinctive consumption choices. The PEB and NFU variables have been standardized (mean = 0, st.dev = 1).

Table 7 reports ordered logistic regressions examining how these mechanisms relate to changes in stated intention to purchase second-hand clothing. Column (1) includes, in addition to Delta Belief, PEB, treatment indicators (using CO2 as baseline), and their interactions. As expected,

PEB strongly predicts increased intention to adopt second-hand (0.508, $p < 0.01$). Notably, the Water $\times$ PEB interaction is positive and highly significant (0.245, $p < 0.01$), indicating that the Water treatment is particularly effective among pro-environmentally oriented individuals. The corresponding interaction with Energy is not significant, suggesting that this amplification effect is specific to water-based information. This finding confirms that environmental messaging resonates most strongly with individuals already predisposed toward sustainable consumption, and that this resonance is especially pronounced for water-related information.

Table 7: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	(1)	(2)	(3)
Delta Belief	-0.004** (0.002)	-0.002 (0.002)	-0.004** (0.002)
Water	0.340*** (0.056)	0.370*** (0.056)	0.342*** (0.057)
Energy	0.099* (0.053)	0.128** (0.053)	0.098* (0.054)
Pro-Environmental Behavior	0.508*** (0.037)		0.453*** (0.038)
Need for Uniqueness		-0.356*** (0.038)	-0.273*** (0.039)
Water $\times$ Pro-Environmental Behavior	0.245*** (0.055)		0.259*** (0.058)
Energy $\times$ Pro-Environmental Behavior	0.081 (0.054)		0.089 (0.056)
Water $\times$ Need for Uniqueness		-0.009 (0.057)	0.060 (0.059)
Energy $\times$ Need for Uniqueness		0.012 (0.054)	0.023 (0.057)
Cut 1	-2.246*** (0.050)	-2.122*** (0.047)	-2.270*** (0.050)
Cut 2	0.743*** (0.040)	0.734*** (0.039)	0.748*** (0.040)
Observations	7860	7860	7860

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Pro-Environmental Behavior* and *Need for Uniqueness* are standardized scales. Interaction terms between treatment indicators and these scales are included. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column (2) includes NFU, treatment indicators, and their interactions. We find that NFU exhibits a negative and highly significant association with increased intention to buy second-hand (-0.356 , $p < 0.01$). Individuals who score higher on NFU are significantly less likely to report increased willingness to adopt second-hand clothing after learning about environmental benefits. Neither the Water $\times$ NFU nor the Energy $\times$ NFU interactions are significant, indicating that the

NFU barrier operates uniformly across treatment conditions. This counterintuitive result can be understood through two complementary mechanisms. First, high-NFU individuals may paradoxically seek differentiation through conformity to the familiar: while they desire to feel special and unique, they may achieve this psychological goal by doubling down on conventional consumption patterns (first-hand shopping) that they already know and master, rather than venturing into the less familiar territory of second-hand retail. In this interpretation, uniqueness is constructed not through counter-normative behavior but through confident adherence to established consumption modes.

Second, high-NFU individuals may be particularly prone to reactance against perceived experimenter demand. When presented with information explicitly highlighting environmental benefits—which carries an implicit normative message that “you should care about this”—individuals motivated by uniqueness may resist precisely because the desired response seems too obvious or prescribed. This resistance represents a form of asserting autonomy and differentiation: by rejecting the environmentally virtuous option that the experiment appears to endorse, high-NFU respondents signal that they make independent choices unconstrained by external expectations.

Column (3) includes both mechanisms and all interactions simultaneously. The effects of PEB (0.453, $p < 0.01$) and NFU (-0.273 , $p < 0.01$) remain highly significant, confirming that these mechanisms operate independently of each other. The Water $\times$ PEB interaction remains significant (0.259, $p < 0.01$), while none of the NFU interactions reach significance. These results indicate an important asymmetry: while NFU acts as a uniform barrier across all treatment conditions, pro-environmental orientation selectively amplifies the effectiveness of water-based information. Water and Energy remain statistically significant with respect to CO2 throughout the model specifications.

Table 8 and Table 9 extend this analysis to WTA and WTP, allowing us to assess whether these mechanisms influence consequential choices in addition to stated intentions. Focusing first on WTA (Table 8), column (1) includes PEB, treatment indicators (using Control as baseline), and their interactions. We observe that PEB significantly reduces WTA (-2.500 , $p < 0.01$), meaning pro-environmentally oriented individuals require less compensation to switch from first-hand to second-hand vouchers. The Energy $\times$ PEB interaction is marginally significant and positive (1.283, $p < 0.10$), suggesting that the PEB effect on WTA is somewhat attenuated in the Energy condition relative to Control; no other PEB interactions are significant.

Column (2) includes NFU and its interactions. NFU significantly increases WTA (2.211, $p < 0.01$). High-NFU individuals demand greater compensation to switch to second-hand vouchers, reinforcing the pattern observed for stated intentions. This effect suggests that resistance to second-hand adoption among individuals with high NFU extends beyond stated preferences to consequential financial trade-offs. None of the NFU interactions are significant, indicating that this barrier operates uniformly across treatments.

Column (3) includes both mechanisms simultaneously. PEB (-2.166 , $p < 0.01$) and NFU (1.822, $p < 0.01$) both retain their significance and direction, confirming that these mechanisms operate independently. No interaction terms reach significance in this joint specification. As for the stated intention, all treatments remain statistically significant throughout the model specifications.

Turning to WTP (Table 9), the patterns reveal important asymmetries. Column (1) shows that PEB strongly increases WTP (2.037, $p < 0.01$). The positive coefficient suggests that pro-

Table 8: Tobit Regression Estimations of WTA for Second-Hand

	(1)	(2)	(3)
Water	-4.272*** (0.664)	-4.294*** (0.663)	-4.279*** (0.660)
Energy	-3.301*** (0.677)	-3.190*** (0.676)	-3.254*** (0.673)
CO2	-2.735*** (0.675)	-2.513*** (0.674)	-2.691*** (0.671)
Pro-Environmental Behavior	-2.500*** (0.506)		-2.166*** (0.511)
Need for Uniqueness		2.211*** (0.489)	1.822*** (0.493)
Water $\times$ Pro-Environmental Behavior	-0.149 (0.702)		-0.043 (0.720)
Energy $\times$ Pro-Environmental Behavior	1.283* (0.731)		1.193 (0.739)
CO2 $\times$ Pro-Environmental Behavior	0.020 (0.692)		0.119 (0.703)
Water $\times$ Need for Uniqueness		0.042 (0.691)	-0.172 (0.710)
Energy $\times$ Need for Uniqueness		-0.712 (0.700)	-0.498 (0.705)
CO2 $\times$ Need for Uniqueness		0.173 (0.682)	0.120 (0.691)
Constant	7.705*** (0.489)	7.603*** (0.488)	7.671*** (0.486)
Observations	5241	5241	5241

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTA*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Negative values indicate lower WTA (greater willingness to accept second-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control*. *Pro-Environmental Behavior* and *Need for Uniqueness* are standardized scales. Interaction terms between treatment indicators and these scales are included. Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

environmental individuals who have just been informed about environmental benefits find the imposed second-hand baseline more acceptable and thus require a higher premium to switch away from it. In other words, they are not eager to pay for first-hand; rather, they exhibit higher measured WTP because they are relatively resistant to abandoning the second-hand option when it is presented as the default. Notably, the Water $\times$ PEB interaction is positive and significant (1.089, $p < 0.05$), indicating that pro-environmental individuals respond particularly strongly to water-based information also in the WTP task—echoing the interaction pattern found for stated intentions.

Column (2) shows that NFU has no significant association with WTP (-0.217 , $p > 0.10$), in sharp contrast with the strong WTA effect. None of the NFU interactions are significant. This asymmetry is consistent with our interpretation: when first-hand is the natural default (WTA), high-NFU individuals strongly resist switching because doing so would mean conforming to the

Table 9: Tobit Regression Estimations of WTP for First-Hand

	(1)	(2)	(3)
Water	1.317** (0.521)	1.574*** (0.532)	1.319** (0.521)
Energy	0.619 (0.520)	0.762 (0.530)	0.613 (0.521)
CO2	1.023* (0.522)	1.106** (0.532)	1.018* (0.522)
Pro-Environmental Behavior	2.037*** (0.384)		2.117*** (0.396)
Need for Uniqueness		-0.217 (0.382)	0.312 (0.386)
Water $\times$ Pro-Environmental Behavior	1.089** (0.541)		1.131** (0.556)
Energy $\times$ Pro-Environmental Behavior	0.183 (0.558)		0.119 (0.573)
CO2 $\times$ Pro-Environmental Behavior	0.545 (0.539)		0.488 (0.556)
Water $\times$ Need for Uniqueness		0.216 (0.547)	0.327 (0.553)
Energy $\times$ Need for Uniqueness		-0.219 (0.543)	-0.245 (0.549)
CO2 $\times$ Need for Uniqueness		-0.300 (0.542)	-0.215 (0.550)
Constant	-4.076*** (0.367)	-4.206*** (0.373)	-4.071*** (0.367)
Observations	5255	5255	5255

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTP*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Positive values indicate higher WTP (greater willingness to pay for first-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control*. *Pro-Environmental Behavior* and *Need for Uniqueness* are standardized scales. Interaction terms between treatment indicators and these scales are included. Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

experimenter’s apparent endorsement of second-hand shopping. When second-hand is artificially imposed as the default (WTP), the NFU barrier disappears, suggesting that uniqueness motives are activated primarily when individuals are asked to relinquish their natural default rather than when they are offered the opportunity to return to it.

Column (3) confirms these patterns: PEB remains strongly significant (2.117, $p < 0.01$), the Water $\times$ PEB interaction remains significant (1.131, $p < 0.05$), and NFU is again insignificant (0.312, $p > 0.10$). As in the main treatment analysis, Water and CO2 coefficients are significant against the Control.

Taken together, the analyses on stated intentions and voucher choices reveal a consistent picture. PEB operates as a robust amplifier of responsiveness across all outcomes: it increases stated intentions, reduces WTA, and raises WTP. Moreover, its effect is selectively enhanced for water-based information, as the Water $\times$ PEB interaction is significant for stated intentions (Table 7)

and WTP (Table 9). NFU acts as a uniform barrier to second-hand adoption in stated intentions and WTA, but—importantly—this barrier does not extend to the WTP frame, where NFU shows no significant effect. Neither mechanism interacts significantly with treatment type for NFU, confirming that uniqueness motives reduce general receptivity to environmental information regardless of the specific metric used. These patterns are robust to the joint inclusion of both mechanisms (column 3 in all tables).

Result 7 (PEB): *Pro-environmental behavior consistently amplifies responsiveness to environmental information across stated intentions, WTA, and WTP. The Water $\times$ PEB interaction is significant for stated intentions and WTP, indicating that water-based information is particularly effective among pro-environmentally oriented individuals.*

Result 8 (NFU): *Need for Uniqueness operates as a barrier to second-hand adoption in stated intentions and WTA, but not in the WTP frame. This barrier is uniform across treatment conditions, with no significant NFU $\times$ treatment interactions.*

5 Additional Results and Robustness Checks

5.1 Study of Beliefs

To extend the analysis on how information provision affects behavioral outcomes, we analyze the distribution and accuracy of respondents’ prior beliefs about the environmental benefits of second-hand clothing. Understanding the structure of these beliefs is crucial for interpreting our treatment effects, as the magnitude and direction of belief updating may vary substantially across individuals and treatment conditions.

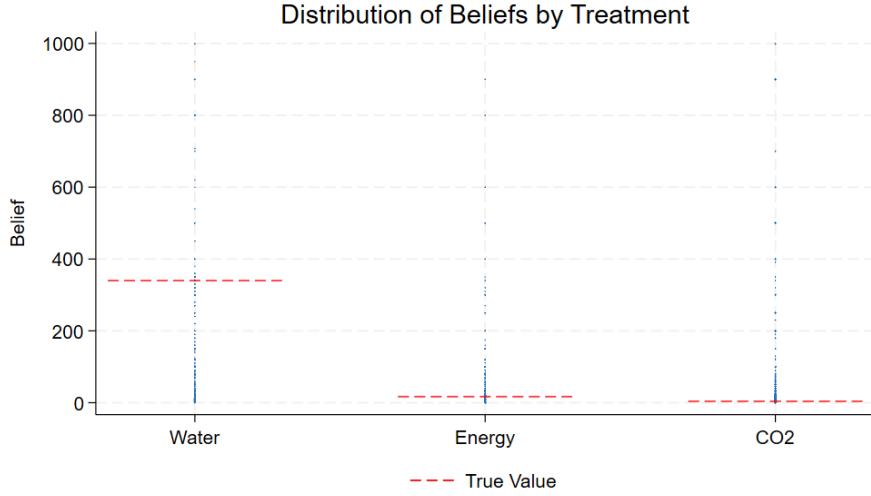
Figure 7 presents scatter plots showing the distribution of respondents’ initial beliefs against the true values for each environmental externality. Several patterns emerge from this visualization. First, there is substantial heterogeneity in the distribution of prior beliefs across all three environmental costs, with respondents’ guesses spanning a wide range around the true values. Second, while a few respondents cluster near the correct values, a notable proportion systematically under- or overestimate the environmental benefits.

This variation in scale warrants careful consideration. While the water treatment begins with an anchor of 10 liters (compared to 1 unit for energy and CO₂), the true value of 340 liters represents a 34-fold increase from the anchor—effectively equivalent to an anchor-normalized value of 34, which is still substantially larger than the other two treatments. This difference in magnitude may influence both the difficulty of the belief elicitation task and the psychological impact of the information provision, as larger absolute numbers may be perceived as more substantial regardless of their real-world equivalence.

Following our pre-registered analysis plan, we identify and examine outliers in the belief data using Tukey’s rule: observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are classified as outliers, where $Q1$ and $Q3$ denote the first and third quartiles, and IQR is the interquartile range. Figure 8 displays violin plots both with and without outliers, allowing visual comparison of how outlier removal affects the distribution of beliefs across treatments.

The impact of outlier removal is not uniform across treatments. To examine the distribution of belief accuracy more systematically, Figure 9 presents violin plots of Delta Belief – the standardized

Figure 7: Distribution of Beliefs vs. True Values by Treatment



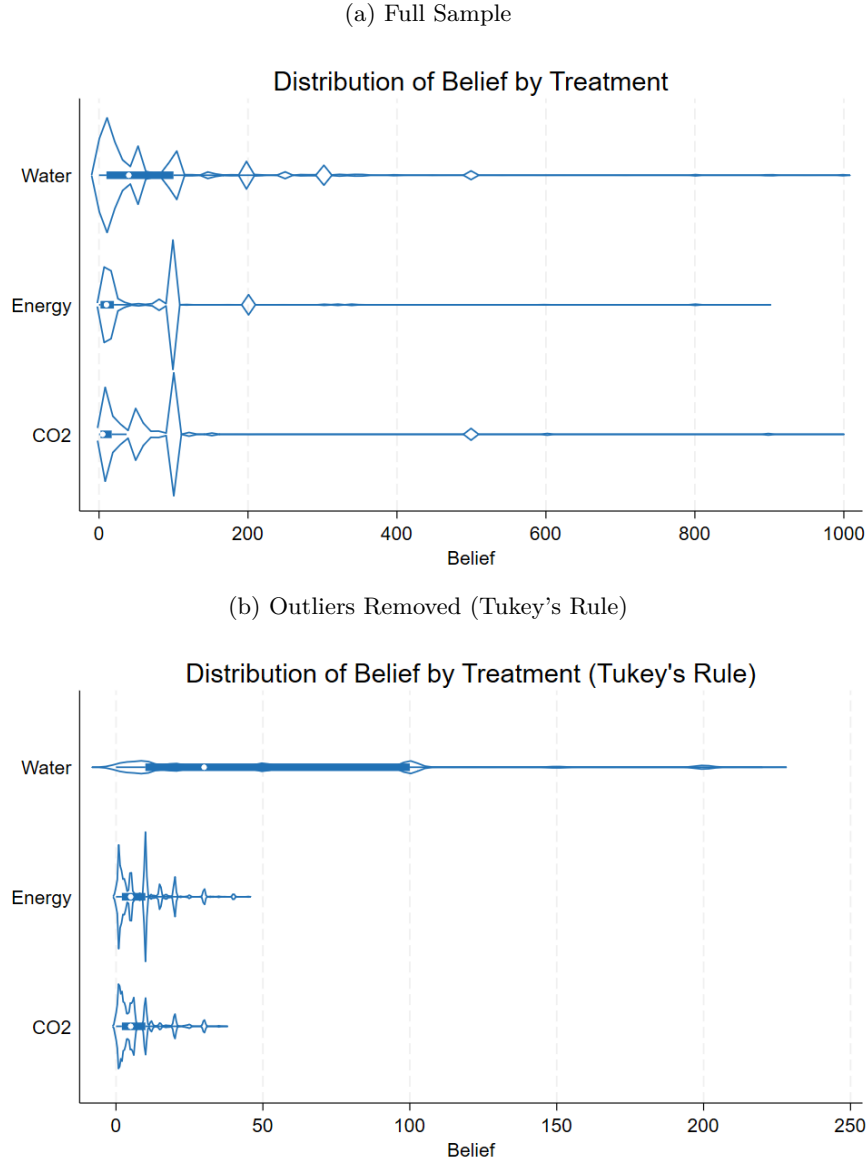
Note: Scatter plots showing respondents' initial beliefs about environmental savings against the true values for each treatment. The dashed line represents perfect accuracy (belief = true value). Each point represents an individual respondent. True values: Water = 340 liters, Energy = 17 kWh, CO2 = 4 kg.

distance between respondents' beliefs and the true values, computed as $(\text{belief} - \text{true value}) / \text{true value}$ – for each treatment. For CO2 and energy, eliminating outliers produces distributions that remain reasonably balanced between under- and overestimation, preserving the essential structure of belief heterogeneity. However, for the water treatment, outlier removal has a larger effect: the resulting distribution consists exclusively of respondents who underestimated the true value. This asymmetry arises because the upper tail of the water belief distribution—those who overestimated water savings—is disproportionately classified as outliers due to the large true value of 340 liters, while the lower tail remains within the non-outlier range.

This differential impact raises an important methodological consideration. While outlier removal is a standard practice and was pre-registered in our analysis plan, its application to the water treatment fundamentally alters the composition of the sample in that condition, leaving primarily underestimators. Such a transformation could artificially amplify treatment effects for water if underestimators respond more strongly to information provision—a pattern we indeed observe in our main results. To address this concern, we report all main analyses both with and without outlier removal. The primary results presented in the main text include all observations to preserve sample composition and comparability across treatments. However, we provide complete robustness analyses excluding outliers in [Appendix B](#), allowing readers to assess the sensitivity of our findings to this specification choice.

Result 9 (Belief heterogeneity): *Respondents' prior beliefs exhibit substantial heterogeneity across all treatments, with the majority systematically underestimating environmental benefits.*

Figure 8: Distribution of Beliefs: Full Sample and Outliers Removed

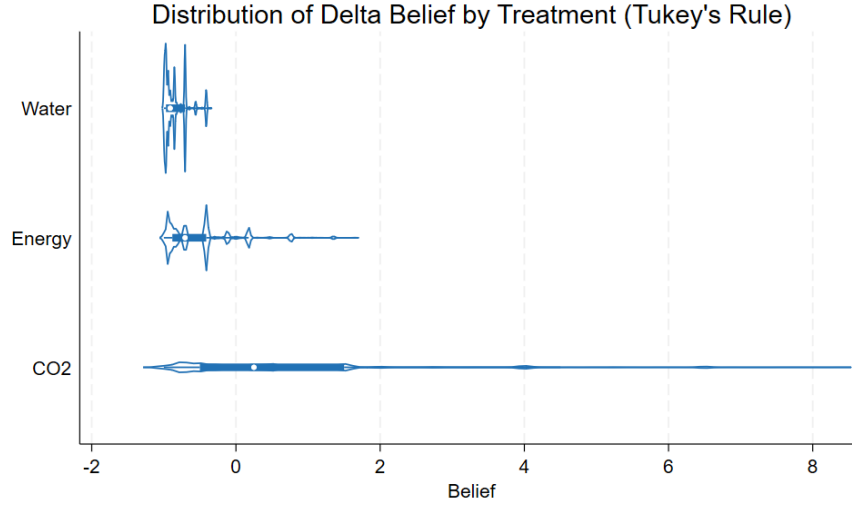


Note: Violin plots showing the distribution of beliefs for each treatment. Panel (a): full sample. Panel (b): after removing outliers using Tukey's rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). Each violin shows the probability density of belief values, with wider sections indicating higher concentration of responses.

5.2 Default Options Across MPL Frames

A key feature of our experimental design is the use of two multiple price list (MPL) frames: one with first-hand (FH) as the fixed option and another with second-hand (SH) as the fixed option. This dual-frame approach allows us to test whether treatment effects are robust to the way choice problems are presented, and whether individuals exhibit consistent preferences when the fixed

Figure 9: Distribution of Delta Belief: Outliers Removed



Note: Violin plots showing the distribution of Delta Belief (standardized distance between belief and true value) for each treatment after removing outliers using Tukey’s rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). Negative values indicate underestimation; positive values indicate overestimation.

option is manipulated.

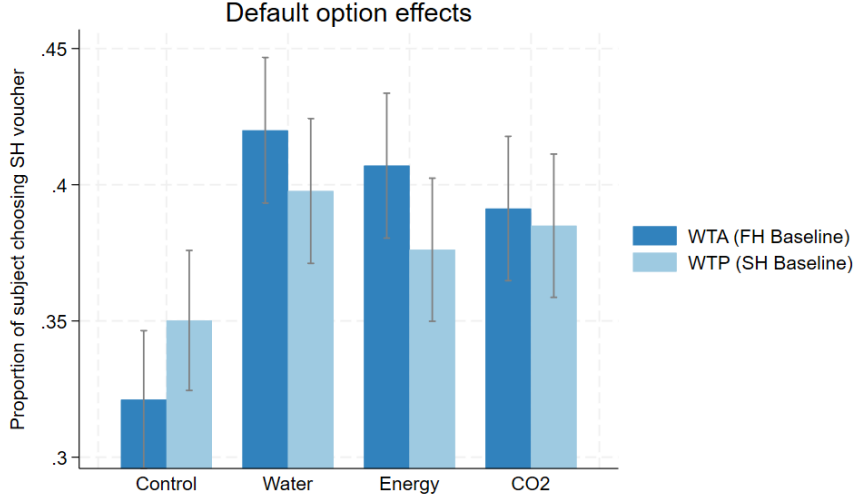
To examine preference consistency across frames, we focus on the critical choice where both vouchers are valued equally at €10—the point at which neither option offers a financial advantage and the decision reflects preference for the type (either FH, or SH) net of its monetary value. For each respondent, we code a binary variable equal to 0 if they choose the first-hand voucher and 1 if they choose the second-hand voucher at this equal-value point, regardless of which option served as the baseline in their assigned MPL. This measure allows us to compare preferences in a financially neutral setting across the two framing conditions.

Figure 10 presents the proportion of respondents choosing second-hand at the €10-€10 point, broken down by MPL baseline (FH vs. SH) and by treatment condition. The results reveal remarkable consistency across framing conditions, with proportion test details reported in Table 10. Pooling across all treatments, 38.5% of respondents with FH as baseline chose second-hand, compared to 37.7% with SH as baseline—a difference of less than one percentage point that is not statistically significant. This suggests that, on average, respondents exhibit stable preferences regardless of which option is framed as the status quo.

Treatment-specific comparisons similarly reveal no significant differences between the two baseline conditions. In the Control condition, 32.1% of respondents with FH baseline chose second-hand compared to 35.0% with SH baseline, a 2.9 percentage point difference that does not reach statistical significance. The Water treatment shows 42.0% versus 39.8%, Energy shows 40.7% versus 37.6%, and CO2 shows 39.1% versus 38.5%. None of these within-treatment comparisons reach conventional levels of statistical significance, indicating that the status quo framing does not systematically alter choice behavior at the point of equal valuation.

These findings provide reassuring evidence that our main results are not driven by framing

Figure 10: Proportion Choosing Second-Hand at Equal Value (€10-€10), by Baseline and Treatment



Note: Bar chart showing the proportion of respondents choosing the second-hand voucher when both options are valued at €10, separately by MPL baseline (FH vs. SH) and treatment condition. Error bars represent 95% confidence intervals.

effects or status quo bias. Despite the substantial literature documenting endowment effects and reference-dependent preferences (Marzilli Ericson and Fuster, 2014; Brown et al., 2024), respondents in our experiment appear to exhibit relatively stable preferences when choosing between equally-valued vouchers, regardless of which option is designated as the baseline. This consistency strengthens the interpretation of our WTA and WTP results: while the magnitude of stated valuations differs across frames (as expected given the WTA-WTP gap), the underlying directional preferences appear robust to manipulations of the status quo.

Table 10: Proportion Tests: Indifference by Voucher Baseline

	WTA (Baseline FH)	WTP (Baseline SH)	Difference	z-stat	p-value
Total	0.385	0.377	−0.008	0.81	0.418
Control	0.321	0.350	0.029	−1.58	0.114
Water	0.420	0.398	−0.022	1.16	0.247
Energy	0.407	0.376	−0.031	1.62	0.106
CO2	0.391	0.385	−0.006	0.33	0.740

Note: Two-sample proportion tests comparing the share of indifferent respondents (switching at the baseline value of €10) across the two elicitation tasks. *WTA (Baseline FH)* reports the proportion of respondents indifferent between the first-hand and second-hand voucher in the WTA task; *WTP (Baseline SH)* reports the corresponding proportion in the WTP task. *Difference* is WTP minus WTA proportion. Tests are reported for the full sample and separately by treatment group.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

That said, it is important to note that this consistency test examines only a single point in

the choice set—the equal-value condition—and does not capture potential framing effects that may operate when the vouchers differ in value. Our earlier WTA and WTP analyses revealed meaningful differences in treatment effects across the two frames, particularly regarding the role of belief accuracy. The present analysis therefore complements rather than contradicts those findings, suggesting that while respondents exhibit consistent preferences at the point of indifference, the path by which they reach that point (i.e., the trade-offs they are willing to make) can vary with the framing of the choice problem.

Result 10 (Default option effect): *Respondents exhibit consistent preferences across MPL frames at the point of equal valuation (€10-€10), with no significant differences in second-hand choice between FH baseline (38.5%) and SH baseline (37.7%) conditions, either overall, or within any individual treatment.*

5.3 Stores proximity

An important consideration when interpreting our results is the role of store accessibility in moderating treatment effects. While information about environmental benefits may shift preferences toward second-hand clothing, the feasibility of acting on these preferences depends critically on whether suitable retail options are available nearby. In this section, we examine whether the effects we observe vary systematically with respondents’ proximity to the stores featured in our voucher task.

Our dataset includes 2,435 unique postal codes corresponding to respondents’ locations across Italy, and we need to identify the closest FH and SH stores from a total of 308 store locations (covering both chains). Computing the precise driving distance between every respondent location and every store would require 749,980 API requests to Google Maps, at substantial computational cost. To make this process tractable while maintaining accuracy, we implemented a multi-stage filtering procedure.

First, we retrieved geographic coordinates for all store addresses and postal codes using the Google Maps Geocoding API. For postal codes associated with multiple municipalities, we computed the geometric center of the area to establish a single reference point. Second, we used the geodesic distance formula to compute straight-line distances between all respondent locations and stores, accounting for the Earth’s curvature. This allowed us to efficiently identify candidate stores within a reasonable distance threshold without yet accounting for road networks and actual travel routes.

We examined several distance thresholds to balance comprehensiveness with feasibility. Setting the threshold at 200 km retained sufficient store options for the vast majority of respondents while reducing the number of required API calls to approximately 25% of the initial total. Further refinement to 140 km reduced the computational burden to just 15% of the original requests (114,370 distance calculations) while still ensuring that nearly all respondents—including those in remote locations such as Sardinia—had at least one store of each type within range. The only exception was respondents on the small islands of Pantelleria and La Maddalena, who were retained in the sample given their negligible number.

Using this filtered set of candidate stores, we then queried the Google Maps Routes API (DistanceMatrix) to obtain precise driving distances accounting for actual road networks. For each

postal code, we identified the single closest FH store and the single closest SH store, as well as which of the two was closer overall. This final dataset allows us to test whether proximity to stores amplifies or attenuates the information provision effects documented in the previous sections.

Table 11 reports Tobit regressions examining how store proximity moderates treatment effects on WTA. Column (1) includes the treatment indicators and the binary indicator for which store type is closer. The results reveal an interesting pattern: respondents for whom the second-hand store is the closest option exhibit significantly higher WTA (1.247, $p < 0.01$), meaning they require greater compensation to switch from first-hand to second-hand vouchers. This finding is counter-intuitive at first glance—one might expect proximity to second-hand stores to facilitate adoption. However, the result is consistent with convenience reinforcing existing preferences: when second-hand stores are farther away, respondents may view them as less viable alternatives and thus be more willing to accept lower-valued vouchers to “try” second-hand shopping, knowing the barrier to actually redeeming the voucher is higher.

Column (2) adds interactions between treatment indicators and the SH Closest dummy. None of the interactions are significant, indicating that relative proximity does not differentially moderate specific treatment effects. The SH Closest main effect also loses significance in this specification.

Column (3) replaces the binary proximity indicator with continuous distance measures for both store types. Neither the distance to first-hand stores nor the distance to second-hand stores shows a significant association with WTA. This suggests that absolute accessibility matters less than relative accessibility—it is the comparison between the two store types, rather than their individual distances, that influences willingness to accept second-hand vouchers.

Column (4) adds interactions between treatment indicators and the continuous distance measures. The Energy $\times$ FH Distance interaction is negative and significant (-0.088 , $p < 0.05$), and the Energy $\times$ SH Distance interaction is positive and marginally significant (0.056 , $p < 0.10$), suggesting that the Energy treatment becomes somewhat more effective when first-hand stores are farther away and less effective when second-hand stores are farther away. No other distance interactions are significant.

Critically, across all four specifications, the treatment effects remain stable (with the exception of CO2 in Column (2)). The consistency of these coefficients across specifications indicates that store proximity does not substantially moderate treatment effects—information provision appears equally effective regardless of store accessibility.

Table 12 presents parallel analyses for WTP, where second-hand serves as the baseline. Column (1) shows that the binary proximity indicator has no significant effect on WTP, contrasting sharply with the WTA results. Column (2) adds interactions with SH Closest; none are significant. Column (3) uses continuous distance measures; neither is significant. Column (4) adds distance interactions; none reach significance.

As with WTA, treatment effects remain stable across specifications, with Water always significant and CO2 only in the specification with no interaction effects. Energy shows no significant effect relative to Control in any specification.

The absence of proximity effects for WTP is consistent with our earlier interpretation that imposing second-hand as the status quo creates an artificial reference point that attenuates behavioral responses. When respondents are forced to consider switching from second-hand to first-hand, the relative accessibility of stores appears irrelevant—the decision is driven primarily by preferences

Table 11: Tobit Regression Estimations of WTA for Second-Hand

	(1)	(2)	(3)	(4)
Water	-3.996*** (0.665)	-4.216*** (1.025)	-4.004*** (0.666)	-3.967*** (0.935)
Energy	-2.459*** (0.676)	-2.826*** (1.018)	-2.466*** (0.676)	-2.271** (0.961)
CO2	-2.667*** (0.677)	-1.629 (1.047)	-2.652*** (0.677)	-2.441*** (0.942)
SH Closest	1.247*** (0.477)	1.432 (0.976)		
Water $\times$ SH Closest		0.382 (1.346)		
Energy $\times$ SH Closest		0.662 (1.361)		
CO2 $\times$ SH Closest		-1.784 (1.372)		
FH Distance (in km)			-0.018 (0.014)	0.022 (0.027)
SH Distance (in km)			0.007 (0.011)	-0.016 (0.022)
Water $\times$ FH Distance (in km)				-0.042 (0.038)
Energy $\times$ FH Distance (in km)				-0.088** (0.039)
CO2 $\times$ FH Distance (in km)				-0.032 (0.039)
Water $\times$ SH Distance (in km)				0.028 (0.030)
Energy $\times$ SH Distance (in km)				0.056* (0.033)
CO2 $\times$ SH Distance (in km)				0.013 (0.032)
Constant	6.834*** (0.562)	6.724*** (0.751)	7.694*** (0.546)	7.583*** (0.670)
Observations	5074	5074	5074	5074

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTA*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Negative values indicate lower WTA (greater willingness to accept second-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control*. *SH Closest* is a dummy equal to 1 if the second-hand store is closer than the first-hand store to the respondent's postal code. *FH Distance* and *SH Distance* denote the driving distance (in km) from the respondent's postal code to the nearest first-hand and second-hand store, respectively. Interaction terms between treatment indicators and store-proximity variables are included.

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

rather than logistical considerations.

These results yield two key insights:

Result 11 (Store proximity on WTA): *While relative store proximity affects baseline WTA*

Table 12: Tobit Regression Estimations of WTP for First-Hand

	(1)	(2)	(3)	(4)
Water	1.571*** (0.533)	1.597* (0.825)	1.569*** (0.533)	1.279* (0.760)
Energy	0.707 (0.529)	0.336 (0.832)	0.700 (0.529)	0.475 (0.749)
CO2	1.103** (0.534)	0.679 (0.819)	1.100** (0.533)	1.090 (0.750)
SH Closest	-0.013 (0.383)	-0.343 (0.757)		
Water $\times$ SH Closest		-0.061 (1.081)		
Energy $\times$ SH Closest		0.637 (1.078)		
CO2 $\times$ SH Closest		0.737 (1.080)		
FH Distance (in km)			-0.003 (0.012)	0.006 (0.023)
SH Distance (in km)			-0.008 (0.009)	-0.020 (0.017)
Water $\times$ FH Distance (in km)				-0.015 (0.034)
Energy $\times$ FH Distance (in km)				-0.000 (0.033)
CO2 $\times$ FH Distance (in km)				-0.018 (0.032)
Water $\times$ SH Distance (in km)				0.024 (0.025)
Energy $\times$ SH Distance (in km)				0.010 (0.023)
CO2 $\times$ SH Distance (in km)				0.013 (0.026)
Constant	-4.075*** (0.438)	-3.881*** (0.587)	-3.860*** (0.419)	-3.734*** (0.518)
Observations	5105	5105	5105	5105

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTP*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Positive values indicate higher WTP (greater willingness to pay for first-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control*. *SH Closest* is a dummy equal to 1 if the second-hand store is closer than the first-hand store to the respondent's postal code. *FH Distance* and *SH Distance* denote the driving distance (in km) from the respondent's postal code to the nearest first-hand and second-hand store, respectively. Interaction terms between treatment indicators and store-proximity variables are included.

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(with closer second-hand stores paradoxically increasing WTA), it does not substantially moderate treatment effects. Notably, there are mild interaction effects of store proximity and Energy.

Result 12 (Store proximity on WTP): Store proximity does not influence, nor moderate

treatment effects, in WTP.

5.4 Geographic Heterogeneity: Regional Variation in Treatment Effects

Our analysis thus far has examined treatment effects pooling across respondents of diverse geographic areas. However, the effectiveness of different types of environmental information may vary systematically across Italy’s diverse geographic and socio-economic contexts. Regional differences in economic development, urbanization, educational attainment, and exposure to specific environmental challenges may shape how individuals respond to information about water, energy, and CO2 savings.

To explore potential geographic heterogeneity, we replicate our main stated intention and voucher choice analyses ([Table 2](#), [Table 4](#) and [Table 6](#)) separately for Italy’s five macro-regions: Northwest (Piedmont, Lombardy, Liguria, Valle d’Aosta), Northeast (Veneto, Trentino-Alto Adige, Friuli-Venezia Giulia, Emilia-Romagna), Center (Tuscany, Umbria, Marche, Lazio), South (Abruzzo, Molise, Campania, Puglia, Basilicata, Calabria), and Islands (Sicily, Sardinia). These macro-regional classifications, while broad, capture meaningful variation in climate, infrastructure, economic conditions, and cultural norms. Complete regression results are reported in [Appendix B](#).

Despite the limitations in statistical power, several suggestive patterns emerge from the regional analysis that warrant discussion.

Starting with stated intentions ([Table B.9](#)), the Water treatment significantly outperforms CO2 in all macro-regions except the Islands (the macro-region with the smallest sample size), with the Center showing the largest coefficient (0.676, $p < 0.01$). The Energy treatment is significant only in the Center (0.250, $p < 0.05$) and South (0.200, $p < 0.10$), while showing no advantage over CO2 in the Northwest, Northeast, or Islands. This pattern confirms that the superiority of water-based information is geographically robust, while the relative effectiveness of energy versus CO2 messaging is more regionally specific.

Turning to WTA ([Table B.10](#)), the Water treatment demonstrates remarkable consistency across geographic contexts, showing significant effects in all macro-regions in the baseline specification (columns 1, 3, 5, 7, 9), with coefficients ranging from -3.28 in the Northwest to -5.52 in the Center. This robustness suggests that water-related environmental information resonates broadly across Italy’s diverse regional contexts.

An interesting regional specificity appears in the Islands macro-region, where the Energy treatment exhibits a strong effect on WTA (7.21, $p < 0.01$ in column 9), nearly double the magnitude observed in other regions. Even after controlling for belief updating (column 10), the Energy effect remains substantial and marginally significant. This outsized response may reflect the particular salience of energy issues in Sardinia and Sicily, where geographic isolation amplifies energy costs: islands face unique challenges in energy infrastructure and grid connectivity, and energy poverty rates tend to be higher than in mainland regions. For residents of these areas, information about energy savings from second-hand consumption may connect more directly to lived economic realities and everyday concerns about energy affordability (more on this topic in [Section 5.5](#)).

Northeast and Center regions emerges as the most uniformly responsive area, with all three information treatments showing significant effects in the baseline specification. Conversely, the South shows a more selective pattern: while Water maintains strong effects, neither Energy nor

CO2 reach significance.

Turning to WTP (Table B.11), the regional patterns are different and generally much weaker. Notable exceptions occur in the South, where water is the only treatment showing a robust positive effect on WTP (3.14, $p < 0.01$ in column 7), though this effect weakens when controlling for belief updating (2.203, $p < 0.10$ in column 8); and in the Northwest, where CO2 shows significance (2.03, $p < 0.05$). Elsewhere, Water and CO2 are not significant, and Energy fails to reach significance in any region.

Several important caveats apply to these regional findings. The sample sizes for individual macro-regions range from 428 to 2,103 observations across the different specifications, substantially reducing power to detect effects and inflating standard errors. Regional classifications necessarily obscure enormous within-region heterogeneity. Moreover, we did not pre-register hypotheses about regional moderation, and the large number of comparisons (5 regions $\times$ 8 treatments) raises concerns about multiple testing and the possibility of spurious findings. We therefore emphasize that these results should be interpreted as hypothesis-generating rather than hypothesis-testing, pointing toward promising directions for future research with larger, regionally-stratified samples designed explicitly to test geographic heterogeneity.

Nevertheless, the consistency of certain patterns suggests that local environmental conditions and economic contexts may meaningfully shape how individuals respond to different types of environmental information. In the next section, we investigate whether these regional patterns can be explained by objective indicators (e.g., water scarcity indices, energy poverty rates, air quality measurements) or by subjective perceptions and cultural norms around environmental issues.

5.5 Local Environmental Conditions as Treatment Moderators

While the preceding regional analysis provides a coarse-grained assessment of geographic heterogeneity, it does not directly test whether specific, measurable environmental conditions moderate treatment effects. In this section, we adopt a complementary approach by linking individual-level responses to continuous measures of local environmental challenges and pro-environmental political attitudes. This strategy allows us to examine whether treatment effects vary systematically with the intensity of locally-experienced environmental problems, providing a more direct test of the hypothesis that information resonates most strongly when it connects to salient, lived experiences.

We leverage the geographic granularity of our data—which includes respondents’ region, province, and postal code—to link survey responses with administrative data from Italian national and regional statistical agencies. This approach offers several advantages over the macro-regional analysis. First, it exploits continuous variation in environmental conditions rather than imposing arbitrary geographic boundaries, increasing precision and power. Second, it tests specific, theoretically-motivated mechanisms linking local context to information effectiveness, rather than relying on broad regional labels as proxies for unmeasured contextual factors. Third, it allows us to examine whether different types of environmental information are differentially effective depending on locally-salient challenges.

We construct four categories of moderating variables based on publicly available administrative data from Italian national and regional statistical agencies:

Proxy for exposure to pro-environmental attitudes. We measure the prevalence of environmental

concern in respondents’ local communities using the percentage of votes cast for the green party (Alleanza Verdi e Sinistra) in the 2024 European Parliament elections, aggregated at the provincial level. Data are sourced from Eligendo, the Italian Ministry of Interior’s official electoral database. This variable captures the degree to which respondents are embedded in communities where environmental issues are politically salient, which may amplify the effectiveness of environmental information provision across all treatment conditions.

Factors enhancing Water information treatment. We construct two variables intended to capture the local salience of water-related environmental issues. First, we measure flooding risk using the percentage of provincial land area classified as subject to High Probability Hazard of Flooding in 2020, as mapped by ISPRA (Istituto Superiore per la Protezione e la Ricerca Ambientale), Italy’s national environmental protection agency. Second, we measure water scarcity using the number of days in 2021 during which domestic water usage was subject to partial suspension at the provincial level, as reported by ISTAT (Istituto Nazionale di Statistica). Both variables proxy for direct experience with water-related environmental challenges, which may heighten responsiveness to information about water savings from second-hand clothing.

Factors enhancing Energy information treatment. We measure exposure to energy affordability challenges using the Energy Poverty Index (Indice di Povertà Energetica) for 2020, constructed by ISTAT in collaboration with OIPE (Osservatorio Italiano sulla Povertà Energetica) and available at the regional level. This composite index captures the prevalence of households unable to adequately heat their homes or pay energy bills, reflecting both economic vulnerability and heightened awareness of energy consumption. Respondents in regions with higher energy poverty may be more attentive to information about energy savings.

Factors enhancing CO2 information treatment. We measure air quality using average annual concentrations of four key pollutants in 2021 at the provincial level: nitrogen dioxide (NO₂), ozone (O₃), particulate matter PM10, and particulate matter PM2.5. Data are sourced from ISPRA’s national air quality monitoring network. These pollutants are closely associated with fossil fuel combustion and industrial activity, making them plausible proxies for local exposure to CO₂-generating processes. Respondents in provinces with poorer air quality may find CO₂-related information more personally relevant.

Table 13 reports descriptive statistics for these variables across our sample, revealing substantial geographic variation. For instance, green voting ranges from 2.58% to 15.18% across provinces, flooding risk from 0 to 23.91 days, and the Energy Poverty Index from 3.95 to 15.69 across regions.

The results provide limited evidence that local environmental conditions moderate treatment effects in the hypothesized directions. Table B.12 and Table B.13 report specifications testing whether pro-environmental political attitudes (column 1), water-related challenges (column 2), energy poverty (column 3), and air quality (column 4) amplify the effectiveness of corresponding environmental information treatments.

Green party vote share shows no significant main effect on WTA or WTP, nor do any of the interaction terms with treatment indicators reach significance. This null result suggests that living in a politically pro-environmental province does not systematically amplify responsiveness to environmental information, contrary to the hypothesis that community-level environmental norms would enhance treatment effects. One interpretation is that individual-level pro-environmental attitudes (Result 7) matter more than aggregate political context, or that provincial-level vote

Table 13: Descriptive Statistics: Environmental Context Variables

Variable	N	Mean	SD	Min	Max
Green voting (% , provincial)	10,496	6.02	1.81	2.58	15.18
Flooding risk (% land area, provincial)	10,496	5.50	5.07	0	23.91
Water scarcity (dummy, provincial)	10,425	.13	.34	0	1
Energy Poverty Index (regional)	10,496	7.2	3.84	3.95	15.69
NO ₂ concentration ($\mu\text{g}/\text{m}^3$, provincial)	10,466	23.52	4.41	10	35.6
O ₃ concentration ($\mu\text{g}/\text{m}^3$, provincial)	10,214	59.43	7.81	44	95
PM10 concentration ($\mu\text{g}/\text{m}^3$, provincial)	10,466	18.79	5.14	4	33.31
PM2.5 concentration ($\mu\text{g}/\text{m}^3$, provincial)	10,280	14.49	4.20	7	25

Note: Descriptive statistics for environmental context variables linked to respondents based on their province or region of residence. *Green voting* measured as the vote share for green parties in the 2024 EU elections; *Flooding risk* as the percentage of provincial land area at risk (2020–2021); *Water scarcity* as a dummy equal to 1 if the province declared a water emergency in 2021; *Energy Poverty Index* at the regional level (2020); air quality variables (*NO₂*, *O₃*, *PM10*, *PM2.5*) as average annual concentrations at the provincial level (2020–2021).

shares are too coarse to capture meaningful variation in local environmental culture.

Neither flooding risk nor water scarcity shows the expected positive interaction with the Water treatment. The flooding risk variable (HPH) shows no significant main effect or interaction for WTA or WTP. Interestingly, provinces with water usage restrictions show lower baseline WTP (-2.094 , $p < 0.01$), though the corresponding WTA coefficient (-1.381) does not reach significance. However, the Water treatment interaction with water scarcity, while positive, does not reach conventional significance for WTA and is only marginally significant for WTP (2.734 , $p < 0.10$). This marginal WTP interaction suggests that in water-scarce provinces, the Water treatment may slightly amplify willingness to pay for first-hand options when second-hand is imposed as the status quo—a pattern difficult to interpret substantively and possibly reflecting statistical noise given the multiple tests conducted.

The Energy Poverty Index shows no significant main effect for either WTA or WTP. However, the Energy $\times$ Modified LIHC interaction is negative and marginally significant for WTA (-0.261 , $p < 0.10$), tentatively suggesting that the Energy treatment may be somewhat more effective in regions with higher energy poverty. This effect does not appear for WTP, and given the multiple comparisons involved, should be interpreted cautiously.

For the air quality specifications (column 4), several notable patterns emerge for WTA. NO₂ concentration shows a positive and significant main effect (0.206 , $p < 0.01$), and PM2.5 shows a negative and significant main effect (-0.342 , $p < 0.01$), indicating that air quality indicators are associated with baseline WTA levels. The CO₂ $\times$ O₃ interaction is positive and significant (0.173 , $p < 0.05$), but this suggests that higher ozone concentrations make the CO₂ treatment less effective (higher WTA), opposite the hypothesized direction. For WTP, none of the air quality interactions approach significance.

Contrary to our expectations, treatment effects do not vary systematically with local environmental conditions in ways that would suggest information resonates more strongly when connected to lived environmental challenges. Several explanations are plausible. First, the measures of local environmental conditions, while objectively meaningful, may not map onto subjective salience—in-

dividuals may not be aware of or attentive to provincial-level statistics on flooding risk or air quality. Second, the geographic aggregation levels (provincial for most variables, regional for energy poverty) may obscure finer-grained variation that matters for individual responses. Third, treatment effects may be sufficiently strong and uniform that local contextual moderation is simply not detectable with our sample size and measurement precision. Finally, environmental information may operate through cognitive and affective pathways (belief updating, emotional responses to environmental harm) that are relatively independent of whether the specific externality is locally salient.

Result 13: *Local environmental conditions do not systematically moderate treatment effects on voucher choices in the hypothesized directions. There are few marginally significant interactions that emerge (water scarcity $\times$ Water for WTP, Energy Poverty $\times$ Energy for WTA, both $p < 0.10$; CO₂ $\times$ O₃ for WTA, $p < 0.05$ but opposite the expected sign). This pattern suggests that the effectiveness of different environmental framings is not strongly contingent on whether respondents face corresponding local environmental challenges.*

6 Discussion and Conclusion

This paper investigates how different types of environmental information affects individuals’ willingness to adopt second-hand clothing. Using a large, nationally representative sample of Italian consumers ($N = 10,496$), we combine stated intentions with incentivized voucher choices to provide a comprehensive assessment of information framing effects.

Our results can be summarized as follows. Water-based information generates the strongest behavioral responses across all outcomes, increasing stated intention among 43% of respondents (versus 35% for energy and 31% for CO₂) and reducing WTA by approximately 20% (€2.10) relative to control. Energy shows intermediate effectiveness, while CO₂—the dominant metric in sustainability communication—is the least effective frame. These treatment rankings are robust across specifications and outcome measures. Belief accuracy matters for stated intentions, where underestimators respond most strongly, but does not significantly affect incentivized choices once sociodemographic controls are included, suggesting that observable characteristics jointly determine both prior beliefs and behavioral responsiveness. Pro-environmental behavior amplifies treatment effects across all outcomes, with a particularly strong interaction with water-based information for stated intentions and WTP. Need for uniqueness acts as a uniform barrier in stated intentions and WTA but, notably, not in the WTP frame—consistent with uniqueness motives being activated when individuals are asked to relinquish their natural default rather than when offered the opportunity to return to it. WTA and WTP measures yield asymmetric results: treatment effects are substantially attenuated when second-hand is artificially imposed as the status quo, raising concerns about WTP elicitation in contexts where the baseline is not naturally experienced as the default. Finally, neither local environmental conditions (water scarcity, energy poverty, air quality, pro-environmental political attitudes) nor store proximity systematically moderate treatment effects, suggesting that information operates through context-independent cognitive channels rather than through resonance with locally-salient problems.

These findings contribute to several strands of the literature. We extend work on environ-

mental information provision (Jessoe and Rapson, 2014; Allcott and Taubinsky, 2015) by showing that not only whether but *which* environmental externalities are presented critically determines effectiveness—the widespread reliance on CO2 metrics may reflect scientific convention rather than psychological impact. We contribute to the literature on heterogeneous treatment effects by documenting that belief accuracy channels. Our results on pro-environmental behavior and need for uniqueness add to research on individual moderators of pro-environmental interventions, revealing an asymmetry between WTA and WTP that speaks to debates on reference dependence and the WTA-WTP gap. Finally, the null results on contextual moderation challenge context-dependent persuasion models and suggest limits to the intuition that environmental messages resonate more when matched to local problems.

From a policy perspective, our results suggest that shifting from CO2-based to water-based environmental messaging could meaningfully enhance the effectiveness of sustainability campaigns at no additional cost. This recommendation is especially relevant for sectors like fashion and textiles, where water impacts are substantial and intuitively meaningful to consumers. The finding that treatment effects do not vary with local environmental conditions implies that standardized national campaigns can be deployed without costly geographic customization—water-based messaging appears equally effective in water-scarce and water-abundant regions. However, the concentration of effects among already pro-environmentally oriented individuals highlights the limits of information-based interventions. For high-NFU consumers who resist environmental messaging, alternative framings emphasizing individuality and counter-mainstream identity—rather than environmental virtue—may prove more effective. Our methodological findings also caution that WTP estimates imposing counterfactual defaults may underestimate demand for sustainable options, potentially biasing policy evaluations toward underinvestment in green alternatives.

Several limitations should be acknowledged. We do not observe actual purchasing behavior: while the voucher task is consequential, redemption and follow-through remain unobserved, and field experiments tracking real purchases would strengthen external validity. Our local environmental condition measures are aggregated at provincial or regional levels and may not capture individual-level variation in exposure or subjective salience. Finally, our one-shot information provision differs from real-world campaigns involving repeated exposure and social reinforcement; longitudinal designs would illuminate dynamic effects and optimal messaging frequency. Future research should also pursue process-tracing methods to identify why water and energy metrics outperform CO2—whether through stronger emotional responses, easier visualization, or reduced cognitive load—and develop alternative message frames capable of engaging consumer segments that are resistant to environmental appeals.

In sum, this paper demonstrates that the choice of environmental metric matters substantially for the effectiveness of sustainability communication. Water-based information consistently outperforms energy and CO2, across stated intentions and incentivized choices in a large representative sample. Pro-environmental orientation amplifies these effects while need for uniqueness acts as a barrier, and local environmental conditions do not moderate them. For policymakers, the implication is straightforward: reframing environmental information in terms of water rather than CO2 can enhance the behavioral impact of sustainability campaigns promoting second-hand consumption, though reaching beyond already environmentally engaged consumers remains an open challenge requiring complementary strategies.

A Experimental Screens

A.1 Information provision task

Figure A.1: Illustration of the quiz task in CO2 treatment

Before answering the next question, we want to inform you of the value of one kilogram of carbon dioxide emitted (CO2e). 1 kg CO2e = 6 km driven by a medium-sized gasoline car When you buy a second-hand garment, do you know how much CO2 emission you can avoid by not producing a new one? About: _____ kg CO2e
The exact estimate is about 4 kg CO2e, or about 24 km traveled by a medium-sized gasoline car. Based on this information, are you more or less inclined to buy second-hand in your daily life? ☐ I am more inclined ☐ I am equally inclined ☐ I am less inclined

Figure A.2: Illustration of the quiz task in Energy treatment

Before answering the next question, we want to inform you of the value of a kilowatt-hour (kWh) of electricity. 1 kWh = 10 hours of use of a 100-watt bulb When you buy a second-hand garment, do you know how much energy waste you can avoid by not producing a new one? About: _____ kWh
The exact estimate is about 17kWh, or about 170 hours of use of a 100-watt bulb. Based on this information, are you more or less inclined to buy second-hand in your daily life? ☐ I am more inclined ☐ I am equally inclined ☐ I am less inclined

Figure A.3: Illustration of the quiz task in Water treatment

Before answering the next question, we want to inform you of the value of ten liters (10ℓ) of water.

10ℓ = 2 minutes of open kitchen tap

When you buy a second-hand garment, do you know how much water waste you can avoid by not producing a new one?
About: _____ ℓ

The exact estimate is about 340 liters of water or about 17 minutes of running tap.

Based on this information, are you more or less inclined to buy second-hand in your daily life?

☐ I am more inclined
☐ I am equally inclined
☐ I am less inclined

A.2 Voucher choice task

Figure A.4: Illustration of the voucher task (WTA)

Below, we present you with five pairs of vouchers of different amount for purchases at two chain stores, the “Mercatino” and “H&M.” The first is a chain of second-hand stores; with the selected voucher, you can buy only clothing. The second is a chain of clothing stores. We ask you to indicate, for each pair, which voucher you would prefer to receive. At the end of the data collection, we will randomly draw 500 participants; for each participant drawn, one of the five pairs will be randomly chosen, and he/she will be emailed the voucher selected in that pair.

Voucher A 	Voucher B 	Your choice	
€ 10	€ 0	☐ Voucher A	☐ Voucher B
€ 10	€ 5	☐ Voucher A	☐ Voucher B
€ 10	€ 10	☐ Voucher A	☐ Voucher B
€ 10	€ 15	☐ Voucher A	☐ Voucher B
€ 10	€ 20	☐ Voucher A	☐ Voucher B

Figure A.5: Illustration of the voucher task (WTP)

Below, we present you with five pairs of vouchers of different amount for purchases at two chain stores, the “Mercatino” and “H&M.” The first is a chain of second-hand stores; with the selected voucher, you can buy only clothing. The second is a chain of clothing stores. We ask you to indicate, for each pair, which voucher you would prefer to receive. At the end of the data collection, we will randomly draw 500 participants; for each participant drawn, one of the five pairs will be randomly chosen, and he/she will be emailed the voucher selected in that pair.

Voucher A	Voucher B	Your choice	
			
€ 10	€ 0	☐ Voucher A	☐ Voucher B
€ 10	€ 5	☐ Voucher A	☐ Voucher B
€ 10	€ 10	☐ Voucher A	☐ Voucher B
€ 10	€ 15	☐ Voucher A	☐ Voucher B
€ 10	€ 20	☐ Voucher A	☐ Voucher B

B Additional Tables and Figures

B.1 Balance Table

Table B.1: Balance Table

	WTA CO2	WTA Energy	WTA Water	WTA Control	WTP CO2	WTP Energy	WTP Water	WTP Control	Joint F-test
<i>Mean</i>									
Age	49.00	49.83	49.66	50.12	48.94	49.74	50.61	49.50	0.120
Pro-Environmental Behavior	-0.05	0.02	0.03	0.02	-0.02	0.01	0.04	-0.06	0.083*
Need for Uniqueness	0.00	-0.02	0.01	0.01	0.01	0.01	-0.03	-0.01	0.934
<i>Proportion</i>									
<i>Gender</i>									
Woman	0.509	0.517	0.537	0.480	0.500	0.542	0.508	0.501	0.033**
Man	0.491	0.483	0.463	0.520	0.500	0.458	0.492	0.499	0.033**
<i>Macro-region</i>									
Northwest	0.279	0.259	0.273	0.291	0.260	0.273	0.262	0.259	0.520
Northeast	0.200	0.198	0.187	0.182	0.210	0.206	0.195	0.203	0.647
Central	0.184	0.200	0.213	0.191	0.200	0.183	0.207	0.192	0.466
South	0.225	0.233	0.213	0.230	0.216	0.230	0.231	0.243	0.677
Islands	0.113	0.109	0.115	0.107	0.114	0.108	0.105	0.103	0.968
<i>Education</i>									
Middle School (or lower)	0.074	0.101	0.083	0.088	0.074	0.076	0.091	0.106	0.014**
High School	0.577	0.554	0.532	0.552	0.534	0.550	0.529	0.536	0.220
University	0.309	0.294	0.332	0.308	0.339	0.319	0.326	0.312	0.236
PhD	0.041	0.050	0.053	0.052	0.053	0.056	0.054	0.047	0.727

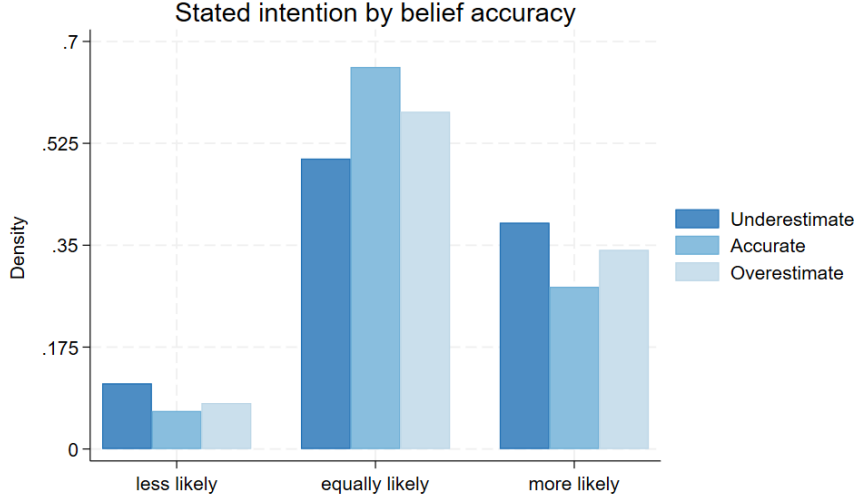
<i>Employment status</i>									
Worker (blue collar)	0.074	0.089	0.072	0.072	0.069	0.073	0.077	0.073	0.660
Employed	0.433	0.366	0.420	0.426	0.424	0.423	0.375	0.384	<.001***
Self-employed	0.116	0.112	0.118	0.102	0.125	0.107	0.124	0.131	0.324
Student	0.054	0.064	0.062	0.056	0.064	0.059	0.056	0.075	0.368
Homemaker	0.094	0.093	0.075	0.085	0.084	0.101	0.079	0.097	0.245
Retired	0.155	0.192	0.178	0.189	0.164	0.170	0.216	0.183	0.002***
Looking for a job	0.054	0.074	0.056	0.056	0.056	0.044	0.052	0.039	0.008***
Other	0.021	0.010	0.018	0.013	0.014	0.024	0.021	0.018	0.112
<i>Income</i>									
Don't know/Prefer not to answer	0.182	0.176	0.180	0.156	0.181	0.196	0.170	0.178	0.321
Less than €1,000	0.042	0.060	0.047	0.048	0.039	0.037	0.040	0.047	0.112
From €1,000 to €1,500	0.119	0.103	0.108	0.114	0.095	0.102	0.110	0.114	0.566
From €1,500 to €2,000	0.165	0.160	0.165	0.175	0.160	0.164	0.161	0.160	0.970
From €2,000 to €3,000	0.208	0.242	0.234	0.230	0.244	0.224	0.238	0.223	0.428
From €3,000 to €5,000	0.184	0.171	0.182	0.185	0.196	0.188	0.188	0.192	0.865
From €5,000 to €10,000	0.060	0.058	0.053	0.059	0.052	0.047	0.066	0.055	0.559
Over €10,000	0.040	0.030	0.030	0.034	0.033	0.043	0.028	0.031	0.331

Note: Balance table reporting sample means (for continuous variables) and proportions (for categorical variables) across treatment arms and elicitation tasks (WTA and WTP). Each row reports group-level statistics for the eight treatment–task cells (CO2, Energy, Water, Control $\times$ WTA, WTP). The last column reports the p -value from a joint F -test of equality of means across all eight groups.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Intention by belief accuracy

Figure B.1: Intention by belief accuracy



Note: Histogram showing the distribution of stated intention changes (less likely, equally likely, more likely to adopt second-hand) across belief accuracy categories (Underestimate, Accurate, Overestimate). Belief accuracy determined using $\pm 15\%$ tolerance interval around true values. Density represents the proportion of respondents in each intention category within each belief accuracy group.

B.3 Intention by treatment

Table B.2: Intention Distribution by Treatment

Intention	Treatment		
	Water	Energy	CO2
Less likely	284 (10.85%)	249 (9.50%)	239 (9.11%)
Equally likely	1,207 (46.12%)	1,447 (55.23%)	1,578 (60.16%)
More likely	1,126 (43.03%)	924 (35.27%)	806 (30.73%)
Total	2,617 (100%)	2,620 (100%)	2,623 (100%)

Note: Cross-tabulation of stated intention changes (less likely, equally likely, more likely to adopt second-hand) across treatment conditions (Water, Energy, CO2). Percentages represent column proportions within each treatment condition. Pearson $\chi^2(4) = 109.56$, $p < 0.001$.

B.4 Regression analysis: Belief outliers

All analyses reported in this section exclude outliers in belief values, identified using Tukey's rule (observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$).

Table B.3: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	(1)	(2)	(3)
Delta Belief	-0.025 (0.017)	0.050** (0.020)	-0.019 (0.022)
Water		0.476*** (0.066)	0.357*** (0.070)
Energy		0.186*** (0.061)	0.117* (0.064)
Need for Uniqueness			-0.145*** (0.028)
Pro-Environmental Behavior			0.491*** (0.030)
Age			-0.008*** (0.002)
Man			0.034 (0.053)
Don't know/Prefer not to answer			-0.473*** (0.078)
Less than €1,000			-0.674*** (0.132)
From €1,000 to €1,500			-0.239** (0.096)
From €1,500 to €2,000			-0.181** (0.081)
From €3,000 to €5,000			-0.054 (0.077)
From €5,000 to €10,000			-0.364*** (0.135)
Over €10,000			0.207 (0.171)
Middle School (or lower)			-0.100 (0.101)

University				-0.049 (0.059)
PhD				-0.219* (0.133)
Worker (blue collar)				-0.202** (0.097)
Self-employed				-0.034 (0.085)
Student				-0.278** (0.124)
Homemaker				-0.170* (0.098)
Retired				-0.358*** (0.094)
Looking for a job				-0.314** (0.125)
Other				-0.106 (0.195)
Yes, I have already purchased them				1.382*** (0.062)
Yes, but I have never purchased them				1.115*** (0.071)
Yes, but I would not repurchase them				0.712*** (0.100)
Northeast				0.066 (0.071)
Central				0.221*** (0.073)
South				0.164** (0.073)
Islands				-0.144 (0.092)
Cut 1	-2.167*** (0.040)	-1.974*** (0.049)	-2.352*** (0.153)	
Cut 2	0.553***	0.763***	0.889***	

	(0.025)	(0.043)	(0.151)
Observations	6698	6698	6698

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Sample excludes outliers in stated beliefs, identified using Tukey’s interquartile range rule. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely). *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, and macro-region of residence (baseline categories set to the mode).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Tobit Regression Estimations of WTA for Second-Hand and WTP for First-Hand

	WTA		WTP	
	(1)	(2)	(3)	(4)
Water	-3.632*** (0.646)	-1.505* (0.778)	1.059** (0.510)	0.602 (0.630)
Energy	-2.168*** (0.666)	0.001 (0.757)	0.558 (0.523)	-0.008 (0.603)
CO2	-2.208*** (0.664)		0.895* (0.519)	
Delta Belief		-0.048 (0.244)		0.335 (0.209)
Need for Uniqueness	0.899*** (0.262)	0.874*** (0.306)	0.660*** (0.202)	0.638*** (0.241)
Pro-Environmental Behavior	-1.487*** (0.273)	-1.422*** (0.313)	1.679*** (0.215)	1.791*** (0.254)
SH Closest	1.223** (0.582)	0.790 (0.668)	0.095 (0.459)	0.246 (0.546)
FH Distance (in km)	-0.010 (0.016)	-0.030 (0.019)	-0.004 (0.014)	-0.007 (0.017)
SH Distance (in km)	-0.013 (0.014)	-0.002 (0.016)	0.000 (0.011)	0.005 (0.013)
Age	0.031 (0.023)	0.022 (0.026)	-0.018 (0.018)	-0.010 (0.021)
Man	-2.580*** (0.494)	-2.687*** (0.567)	1.020** (0.400)	0.710 (0.475)
Don't know/Prefer not to answer	0.705 (0.771)	1.203 (0.887)	-2.022*** (0.602)	-2.033*** (0.722)
Less than €1,000	1.291 (1.279)	1.813 (1.426)	-0.278 (1.044)	0.808 (1.266)
From €1,000 to €1,500	-0.255 (0.881)	-0.049 (1.023)	-1.653** (0.738)	-1.821** (0.883)
From €1,500 to €2,000	-0.170 (0.764)	0.554 (0.903)	-0.157 (0.602)	-0.247 (0.711)
From €3,000 to €5,000	0.084 (0.731)	0.209 (0.853)	-0.547 (0.544)	-0.308 (0.646)

From €5,000 to €10,000	-0.046 (1.140)	0.292 (1.342)	0.464 (0.925)	0.752 (1.104)
Over €10,000	-0.948 (1.385)	-1.712 (1.561)	-0.052 (1.183)	-0.045 (1.430)
Middle School (or lower)	-0.785 (0.951)	-1.017 (1.098)	0.398 (0.745)	-0.242 (0.922)
University	-0.109 (0.556)	0.115 (0.649)	-0.441 (0.443)	-0.706 (0.530)
PhD	-2.477** (1.183)	-3.701*** (1.376)	-0.347 (0.876)	-1.205 (1.009)
Worker (blue collar)	-0.908 (0.880)	-1.609 (0.980)	1.689** (0.768)	1.383 (0.923)
Self-employed	0.117 (0.826)	0.002 (0.956)	1.847*** (0.602)	2.605*** (0.737)
Student	-1.337 (1.070)	-2.283* (1.231)	2.320*** (0.877)	1.860* (1.101)
Homemaker	-0.873 (1.007)	-1.340 (1.183)	0.060 (0.790)	0.265 (0.938)
Retired	-0.464 (0.892)	-0.057 (1.033)	1.453** (0.707)	1.913** (0.837)
Looking for a job	-1.266 (1.088)	-0.535 (1.260)	-0.322 (1.052)	-1.327 (1.226)
Other	-0.813 (2.073)	-0.139 (2.371)	5.062*** (1.575)	3.634* (1.930)
Yes, I have already purchased them	-12.222*** (0.597)	-12.443*** (0.688)	9.619*** (0.471)	9.774*** (0.563)
Yes, but I have never purchased them	-9.291*** (0.668)	-10.273*** (0.778)	7.257*** (0.524)	7.409*** (0.621)
Yes, but I would not repurchase them	-7.264*** (0.880)	-6.962*** (1.014)	6.436*** (0.707)	6.570*** (0.822)
Northeast	-0.980 (0.713)	-1.351* (0.811)	-0.046 (0.555)	-0.639 (0.667)
Central	-0.514 (0.686)	-0.827 (0.785)	0.792 (0.561)	0.281 (0.653)
South	-0.310 (0.681)	-0.225 (0.812)	-0.928* (0.546)	-1.025 (0.644)

Islands	1.425 (0.923)	1.613 (1.065)	-1.045 (0.755)	-1.415 (0.913)
Constant	14.368*** (1.521)	13.065*** (1.780)	-8.833*** (1.162)	-8.305*** (1.367)
Controls	Yes	Yes	Yes	Yes
Observations	4517	3231	4534	3246

Note: Tobit regression estimation with standard errors clustered at the individual level. Sample excludes outliers in stated beliefs, identified using Tukey’s interquartile range rule. Dependent variables: *WTA* (columns 1–2) and *WTP* (columns 3–4), both ranging between –€12.5 and €12.5, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus €10). For *WTA*, negative values indicate lower *WTA* (greater willingness to accept second-hand); for *WTP*, positive values indicate higher *WTP* (greater willingness to pay for first-hand). Regressors. *Water* and *Energy* are treatment indicators. Baseline: *Control* in columns (1) and (3); *CO2* in columns (2) and (4). *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value. *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, macro-region of residence, distance to the nearest first-hand and second-hand store, and whether the closest store is second-hand (baseline categories set to the mode).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.5 Regression analysis: Time outliers

All analyses reported in this section exclude outliers in response times, identified using Tukey's rule (observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$).

Table B.5: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	(1)	(2)	(3)
Delta Belief	-0.003*	-0.001	-0.006***
	(0.002)	(0.002)	(0.002)
Water		0.362***	0.372***
		(0.058)	(0.061)
Energy		0.136**	0.129**
		(0.054)	(0.057)
Need for Uniqueness			-0.158***
			(0.026)
Pro-Environmental Behavior			0.494***
			(0.028)
Age			-0.009***
			(0.002)
Man			0.031
			(0.051)
Don't know/Prefer not to answer			-0.426***
			(0.074)
Less than €1,000			-0.596***
			(0.133)
From €1,000 to €1,500			-0.200**
			(0.093)
From €1,500 to €2,000			-0.138*
			(0.077)
From €3,000 to €5,000			-0.014
			(0.074)
From €5,000 to €10,000			-0.325**
			(0.128)
Over €10,000			0.264
			(0.164)
Middle School (or lower)			-0.130
			(0.098)

University				-0.046 (0.057)
PhD				-0.265** (0.128)
Worker (blue collar)				-0.175* (0.096)
Self-employed				-0.029 (0.081)
Student				-0.330*** (0.117)
Homemaker				-0.173* (0.093)
Retired				-0.307*** (0.090)
Looking for a job				-0.277** (0.116)
Other				-0.082 (0.192)
Yes, I have already purchased them				1.311*** (0.059)
Yes, but I have never purchased them				1.039*** (0.070)
Yes, but I would not repurchase them				0.737*** (0.097)
Northeast				0.033 (0.069)
Central				0.250*** (0.070)
South				0.140** (0.070)
Islands				-0.070 (0.087)
Cut 1	-2.225*** (0.040)	-2.066*** (0.048)	-2.375*** (0.147)	
Cut 2	0.572***	0.743***	0.925***	

	(0.025)	(0.040)	(0.145)
Observations	7283	7283	7283

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Sample excludes outliers in survey completion time, identified using Tukey’s interquartile range rule. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely). *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, and macro-region of residence (baseline categories set to the mode).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Tobit Regression Estimations of WTA for Second-Hand and WTP for First-Hand

	WTA		WTP	
	(1)	(2)	(3)	(4)
Water	-3.718*** (0.644)	-1.718*** (0.649)	1.377*** (0.509)	0.075 (0.533)
Energy	-1.954*** (0.656)	0.089 (0.659)	0.992* (0.508)	-0.322 (0.524)
CO2	-2.144*** (0.651)		1.382*** (0.510)	
Delta Belief		-0.013 (0.022)		0.021 (0.017)
Need for Uniqueness	0.803*** (0.252)	0.839*** (0.289)	0.612*** (0.193)	0.596*** (0.225)
Pro-Environmental Behavior	-1.435*** (0.267)	-1.289*** (0.300)	1.577*** (0.209)	1.684*** (0.241)
SH Closest	1.108* (0.568)	0.631 (0.641)	-0.103 (0.448)	-0.007 (0.520)
FH Distance (in km)	-0.015 (0.016)	-0.037** (0.018)	-0.010 (0.013)	-0.012 (0.016)
SH Distance (in km)	-0.006 (0.013)	0.006 (0.015)	0.003 (0.010)	0.006 (0.012)
Age	0.028 (0.022)	0.022 (0.025)	-0.035** (0.017)	-0.035* (0.020)
Man	-2.646*** (0.480)	-2.721*** (0.541)	0.764** (0.386)	0.433 (0.446)
Don't know/Prefer not to answer	0.603 (0.737)	1.009 (0.835)	-2.040*** (0.579)	-2.032*** (0.678)
Less than €1,000	0.678 (1.263)	1.391 (1.392)	-0.347 (1.048)	0.492 (1.249)
From €1,000 to €1,500	0.193 (0.862)	0.622 (0.975)	-1.468** (0.723)	-1.171 (0.851)
From €1,500 to €2,000	0.005 (0.733)	0.670 (0.849)	-0.202 (0.576)	-0.172 (0.661)
From €3,000 to €5,000	0.624 (0.720)	0.944 (0.824)	-0.547 (0.532)	-0.180 (0.616)

From €5,000 to €10,000	0.535 (1.078)	1.167 (1.227)	-0.185 (0.877)	0.042 (1.016)
Over €10,000	-1.060 (1.376)	-2.007 (1.514)	0.421 (1.141)	0.642 (1.348)
Middle School (or lower)	-0.757 (0.926)	-1.006 (1.050)	0.374 (0.729)	-0.189 (0.883)
University	-0.144 (0.541)	-0.097 (0.618)	-0.311 (0.428)	-0.455 (0.501)
PhD	-2.486** (1.132)	-3.509*** (1.297)	-0.190 (0.852)	-0.728 (0.964)
Worker (blue collar)	-0.842 (0.858)	-1.632* (0.942)	1.242 (0.762)	0.765 (0.896)
Self-employed	0.159 (0.806)	-0.089 (0.915)	1.547*** (0.580)	1.930*** (0.688)
Student	-1.288 (1.024)	-1.957* (1.155)	1.669** (0.823)	1.288 (0.989)
Homemaker	-1.386 (0.955)	-2.016* (1.095)	0.185 (0.762)	0.409 (0.885)
Retired	-0.562 (0.871)	-0.157 (0.988)	1.786*** (0.685)	2.261*** (0.791)
Looking for a job	-1.272 (1.050)	-0.808 (1.194)	0.057 (1.000)	-0.852 (1.130)
Other	-1.339 (2.107)	-0.351 (2.399)	4.177*** (1.506)	2.444 (1.780)
Yes, I have already purchased them	-12.388*** (0.580)	-12.798*** (0.656)	9.731*** (0.454)	9.957*** (0.528)
Yes, but I have never purchased them	-9.318*** (0.659)	-10.390*** (0.757)	7.127*** (0.511)	7.369*** (0.594)
Yes, but I would not repurchase them	-7.291*** (0.856)	-7.237*** (0.969)	6.191*** (0.672)	6.400*** (0.765)
Northeast	-1.087 (0.691)	-1.533** (0.774)	-0.139 (0.537)	-0.727 (0.633)
Central	-0.937 (0.665)	-1.406* (0.745)	0.686 (0.539)	0.288 (0.613)
South	-0.346 (0.656)	-0.305 (0.763)	-0.754 (0.532)	-0.906 (0.614)

Islands	0.735 (0.900)	0.505 (1.019)	-0.762 (0.715)	-1.140 (0.839)
Constant	14.498*** (1.477)	13.467*** (1.662)	-7.825*** (1.119)	-6.425*** (1.279)
Controls	Yes	Yes	Yes	Yes
Observations	4710	3505	4757	3539

Note: Tobit regression estimation with standard errors clustered at the individual level. Sample excludes outliers in survey completion time, identified using Tukey’s interquartile range rule. Dependent variables: *WTA* (columns 1–2) and *WTP* (columns 3–4), both ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). For *WTA*, negative values indicate lower *WTA* (greater willingness to accept second-hand); for *WTP*, positive values indicate higher *WTP* (greater willingness to pay for first-hand). Regressors. *Water* and *Energy* are treatment indicators. Baseline: *Control* in columns (1) and (3); *CO2* in columns (2) and (4). *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value. *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, macro-region of residence, distance to the nearest first-hand and second-hand store, and whether the closest store is second-hand (baseline categories set to the mode). Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.6 Regression analysis: Complete specifications

All analyses reported in this section include the full sample with complete model specifications controlling for sociodemographic variables, individual characteristics, and geographic fixed effects, as detailed in the regression tables.

Table B.7: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	(1)	(2)	(3)
Delta Belief	-0.004** (0.002)	-0.001 (0.002)	-0.006*** (0.002)
Water		0.371*** (0.056)	0.367*** (0.059)
Energy		0.135*** (0.052)	0.132** (0.055)
Need for Uniqueness			-0.160*** (0.025)
Pro-Environmental Behavior			0.487*** (0.027)
Age			-0.008*** (0.002)
Man			0.022 (0.049)
Don't know/Prefer not to answer			-0.456*** (0.072)
Less than €1,000			-0.626*** (0.124)
From €1,000 to €1,500			-0.204** (0.089)
From €1,500 to €2,000			-0.152** (0.074)
From €3,000 to €5,000			-0.058 (0.071)
From €5,000 to €10,000			-0.306** (0.123)
Over €10,000			0.190 (0.157)
Middle School (or lower)			-0.129

			(0.092)
University			-0.026
			(0.055)
PhD			-0.233*
			(0.123)
Worker (blue collar)			-0.161*
			(0.091)
Self-employed			-0.039
			(0.078)
Student			-0.296**
			(0.115)
Homemaker			-0.123
			(0.090)
Retired			-0.302***
			(0.087)
Looking for a job			-0.277**
			(0.114)
Other			-0.083
			(0.178)
Yes, I have already purchased them			1.324***
			(0.057)
Yes, but I have never purchased them			1.078***
			(0.067)
Yes, but I would not repurchase them			0.726***
			(0.092)
Northeast			0.061
			(0.066)
Central			0.265***
			(0.067)
South			0.169**
			(0.068)
Islands			-0.068
			(0.084)
Cut 1	-2.226***	-2.065***	-2.327***
	(0.038)	(0.046)	(0.142)
Cut 2	0.553***	0.726***	0.953***

	(0.024)	(0.039)	(0.139)
Observations	7860	7860	7860

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely). *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, and macro-region of residence (baseline categories set to the mode).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8: Tobit Regression Estimations of WTA for Second-Hand and WTP for First-Hand

	WTA		WTP	
	(1)	(2)	(3)	(4)
Water	-3.688*** (0.623)	-1.572** (0.628)	1.293*** (0.495)	0.214 (0.517)
Energy	-2.145*** (0.633)	0.007 (0.635)	0.779 (0.494)	-0.322 (0.509)
CO2	-2.249*** (0.631)		1.196** (0.494)	
Delta Belief		-0.015 (0.022)		0.022 (0.016)
Need for Uniqueness	0.895*** (0.244)	0.877*** (0.279)	0.601*** (0.188)	0.579*** (0.219)
Pro-Environmental Behavior	-1.386*** (0.256)	-1.294*** (0.288)	1.655*** (0.202)	1.745*** (0.233)
SH Closest	1.184** (0.549)	0.791 (0.619)	-0.075 (0.434)	0.019 (0.504)
FH Distance (in km)	-0.010 (0.015)	-0.027 (0.018)	-0.006 (0.013)	-0.008 (0.015)
SH Distance (in km)	-0.008 (0.013)	0.003 (0.015)	-0.001 (0.010)	0.003 (0.012)
Age	0.033 (0.021)	0.028 (0.024)	-0.034** (0.017)	-0.032 (0.019)
Man	-2.602*** (0.463)	-2.691*** (0.523)	0.847** (0.375)	0.558 (0.434)
Don't know/Prefer not to answer	0.367 (0.717)	0.706 (0.807)	-2.032*** (0.566)	-2.031*** (0.662)
Less than €1,000	0.921 (1.208)	1.238 (1.328)	-0.152 (0.995)	0.855 (1.181)
From €1,000 to €1,500	-0.332 (0.827)	-0.183 (0.941)	-1.276* (0.696)	-1.277 (0.813)
From €1,500 to €2,000	-0.293 (0.709)	0.291 (0.815)	-0.053 (0.560)	-0.063 (0.644)
From €3,000 to €5,000	0.295 (0.690)	0.495 (0.789)	-0.490 (0.515)	-0.243 (0.599)

From €5,000 to €10,000	0.524 (1.056)	1.055 (1.208)	-0.142 (0.865)	-0.078 (1.007)
Over €10,000	-1.073 (1.312)	-1.762 (1.455)	0.684 (1.106)	0.907 (1.302)
Middle School (or lower)	-0.739 (0.886)	-0.927 (0.998)	0.383 (0.712)	-0.200 (0.861)
University	-0.203 (0.522)	-0.047 (0.595)	-0.457 (0.415)	-0.698 (0.486)
PhD	-2.603** (1.098)	-3.609*** (1.246)	-0.299 (0.826)	-0.944 (0.935)
Worker (blue collar)	-0.847 (0.830)	-1.451 (0.912)	1.158 (0.738)	0.723 (0.870)
Self-employed	0.354 (0.781)	0.328 (0.887)	1.553*** (0.566)	2.037*** (0.673)
Student	-1.275 (1.003)	-2.040* (1.130)	1.863** (0.806)	1.411 (0.971)
Homemaker	-1.487 (0.922)	-2.069** (1.052)	-0.145 (0.742)	-0.033 (0.859)
Retired	-0.527 (0.836)	-0.197 (0.947)	1.504** (0.666)	1.867** (0.772)
Looking for a job	-1.326 (1.023)	-0.697 (1.166)	-0.358 (0.946)	-1.177 (1.066)
Other	-1.479 (1.975)	-1.099 (2.221)	3.984*** (1.470)	2.297 (1.751)
Yes, I have already purchased them	-12.453*** (0.561)	-12.711*** (0.635)	9.596*** (0.441)	9.745*** (0.514)
Yes, but I have never purchased them	-9.599*** (0.632)	-10.593*** (0.722)	7.143*** (0.496)	7.297*** (0.575)
Yes, but I would not repurchase them	-7.603*** (0.828)	-7.484*** (0.936)	6.169*** (0.658)	6.260*** (0.748)
Northeast	-1.047 (0.668)	-1.360* (0.747)	-0.010 (0.524)	-0.432 (0.616)
Central	-0.792 (0.644)	-1.133 (0.723)	0.891* (0.522)	0.520 (0.595)
South	-0.191 (0.635)	-0.038 (0.736)	-0.828 (0.517)	-0.848 (0.598)

Islands	1.146 (0.870)	1.216 (0.985)	-0.923 (0.701)	-1.194 (0.822)
Constant	14.498*** (1.423)	13.069*** (1.599)	-7.740*** (1.092)	-6.578*** (1.247)
Observations	5074	3788	5105	3817

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variables: *WTA* (columns 1–2) and *WTP* (columns 3–4), both ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). For *WTA*, negative values indicate lower *WTA* (greater willingness to accept second-hand); for *WTP*, positive values indicate higher *WTP* (greater willingness to pay for first-hand). Regressors. *Water* and *Energy* are treatment indicators. Baseline: *Control* in columns (1) and (3); *CO2* in columns (2) and (4). *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value. *Demographic controls* include age, gender, need for uniqueness score, pro-environmental behavior, income, education, employment status, past adoption of second-hand, macro-region of residence, distance to the nearest first-hand and second-hand store, and whether the closest store is second-hand (baseline categories set to the mode). Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.7 Regression analysis: Macroregions

Table B.9: Ordered Logistic Regression on Change in Intention to Purchase Second-Hand

	North West (1)	North East (2)	Center (3)	South (4)	Islands (5)
Delta Belief	-0.008*** (0.003)	-0.003 (0.004)	0.004 (0.003)	0.000 (0.003)	0.008 (0.006)
Water	0.308*** (0.109)	0.406*** (0.130)	0.676*** (0.123)	0.218* (0.116)	0.217 (0.165)
Energy	0.080 (0.103)	0.048 (0.116)	0.250** (0.123)	0.200* (0.110)	0.072 (0.150)
Cut 1	-2.173*** (0.092)	-2.138*** (0.106)	-2.190*** (0.110)	-1.974*** (0.098)	-1.779*** (0.118)
Cut 2	0.825*** (0.076)	0.765*** (0.088)	0.757*** (0.089)	0.536*** (0.083)	0.789*** (0.107)
Observations	2103	1566	1555	1767	869

Note: Ordered logit regression estimation with standard errors clustered at the individual level. Separate estimation by macro-region of residence. Dependent variable. *Change in intention to purchase more second-hand clothes*: categorical variable ranging from -1 to $+1$ (3 categories). Regressors. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value over the true value. *Water* and *Energy* are treatment indicators. Baseline *CO2*. *Cut 1* and *Cut 2* denote the estimated threshold parameters separating the three categories of the ordered dependent variable (less likely / equally likely / more likely).

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.10: Tobit regression estimations of WTA for second-hand

	North West		North East		Center		South		Islands	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Water	-3.279*** (1.254)	-2.134* (1.271)	-4.697*** (1.531)	-1.632 (1.469)	-5.517*** (1.455)	-1.365 (1.449)	-3.829*** (1.419)	-3.180** (1.574)	-4.718** (2.233)	-1.870 (2.252)
Energy	-2.917** (1.295)	-1.617 (1.295)	-3.089** (1.500)	0.002 (1.422)	-2.882* (1.537)	1.263 (1.517)	-2.252 (1.384)	-1.547 (1.531)	-7.212*** (2.350)	-4.286* (2.355)
CO2	-1.779 (1.283)		-3.345** (1.476)		-4.266*** (1.536)		-0.910 (1.457)		-3.275 (2.222)	
Delta Belief		-0.100** (0.040)		-0.046 (0.048)		-0.022 (0.068)		-0.022 (0.036)		-0.070 (0.051)
Constant	7.797*** (0.927)	6.515*** (0.946)	7.317*** (1.127)	4.181*** (1.046)	7.442*** (1.119)	3.255*** (1.105)	6.881*** (0.987)	6.279*** (1.176)	9.811*** (1.645)	6.952*** (1.647)
Observations	1443	1062	1005	766	1032	782	1180	879	581	441

Note: Tobit regression estimation with standard errors clustered at the individual level. Separate estimation by macro-region of residence: North West (columns 1–2), North East (columns 3–4), Center (columns 5–6), South (columns 7–8), Islands (columns 9–10). Dependent variable: *WTA*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Negative values indicate lower WTA (greater willingness to accept second-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control* in odd columns; *CO2* in even columns. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value.

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.11: Tobit regression estimations of WTP for first-hand

	North West		North East		Center		South		Islands	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Water	1.578 (0.985)	-0.371 (0.992)	0.168 (1.155)	-0.372 (1.276)	1.271 (1.212)	1.267 (1.208)	3.139*** (1.134)	2.203* (1.255)	1.290 (1.834)	1.553 (1.862)
Energy	1.442 (0.995)	-0.518 (0.995)	-0.194 (1.068)	-0.779 (1.178)	0.718 (1.193)	0.662 (1.180)	1.172 (1.187)	0.081 (1.290)	-0.098 (1.818)	0.004 (1.830)
CO2	2.030** (1.034)		0.688 (1.105)		0.232 (1.210)		1.361 (1.170)		0.361 (1.657)	
Delta Belief		0.009 (0.022)		0.025 (0.056)		0.030 (0.026)		0.071 (0.059)		0.087** (0.036)
Constant	-4.514*** (0.740)	-2.522*** (0.750)	-3.586*** (0.739)	-3.106*** (0.879)	-3.024*** (0.873)	-2.982*** (0.868)	-5.416*** (0.789)	-4.460*** (0.929)	-4.175*** (1.226)	-4.417*** (1.229)
Observations	1384	1041	1069	800	1028	773	1210	888	564	428

Note: Tobit regression estimation with standard errors clustered at the individual level. Separate estimation by macro-region of residence: North West (columns 1–2), North East (columns 3–4), Center (columns 5–6), South (columns 7–8), Islands (columns 9–10). Dependent variable: *WTP*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Positive values indicate higher WTP (greater willingness to pay for first-hand). Regressors. *Water*, *Energy* and *CO2* are treatment indicators. Baseline: *Control* in odd columns; *CO2* in even columns. *Delta Belief* denotes the standardized distance between the belief and the true value, computed as the difference between the belief and the true value divided by the true value.

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B.8 Regression analysis: Local Environmental Conditions

Table B.12: Tobit Regression Estimations of WTA for Second-Hand

	(1)	(2)	(3)	(4)
Water	-6.433*** (2.365)	-2.607*** (0.890)		
Energy	-3.800 (2.370)		0.948 (1.189)	
CO2	-2.192 (2.378)			-9.560 (7.276)
Green vote share	-0.012 (0.276)			
Water $\times$ Green vote share	0.355 (0.379)			
Energy $\times$ Green vote share	0.093 (0.381)			
CO2 $\times$ Green vote share	-0.056 (0.385)			
HPH		0.054 (0.053)		
Water $\times$ HPH		0.012 (0.114)		
Days of partial suspension (dummy)		-1.381 (0.861)		
Water $\times$ Days of partial suspension (dummy)		2.100 (1.746)		
Modified LIHC			0.059 (0.072)	
Energy $\times$ Modified LIHC			-0.261* (0.149)	
NO2 Concentration				0.206*** (0.073)
CO2 $\times$ NO2 Concentration				-0.086 (0.145)
PM210 Concentration				0.064 (0.124)
CO2 $\times$ PM210 Concentration				-0.199 (0.240)

PM2.5 Concentration				-0.342***
				(0.132)
CO2 × PM2.5 Concentration				0.397
				(0.267)
O3 Concentration				-0.066
				(0.044)
CO2 × O3 Concentration				0.173**
				(0.087)
Constant	7.709***	5.592***	4.932***	8.558**
	(1.711)	(0.453)	(0.582)	(3.655)
Observations	5241	5208	5241	5000

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTA*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Negative values indicate lower *WTA* (greater willingness to accept second-hand). Each column estimates a separate specification including only the treatment relevant to the environmental context variable. Column (1): *Water* treatment interacted with provincial *Green vote share* (2024 EU elections). Column (2): *Water* treatment interacted with provincial *HPH* (hours of partial water suspension) and *Days of partial suspension* (dummy). Column (3): *Energy* treatment interacted with regional *Modified LIHC* (energy poverty index). Column (4): *CO2* treatment interacted with provincial air quality indicators (*NO₂*, *PM10*, *PM2.5*, *O₃* concentrations). Baseline: *Control* in all columns. Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.13: Tobit Regression Estimations of WTP for First-Hand

	(1)	(2)	(3)	(4)
Water	3.691*	0.396		
	(1.913)	(0.704)		
Energy	-0.083		0.913	
	(1.883)		(0.920)	
CO2	1.422			-2.228
	(1.909)			(5.631)
Green vote share	0.363			
	(0.222)			
Water $\times$ Green vote share	-0.358			
	(0.304)			
Energy $\times$ Green vote share	0.133			
	(0.303)			
CO2 $\times$ Green vote share	-0.062			
	(0.308)			
HPH		-0.034		
		(0.043)		
Water $\times$ HPH		0.022		
		(0.086)		
Days of partial suspension (dummy)		-2.094***		
		(0.683)		
Water $\times$ Days of partial suspension (dummy)		2.734*		
		(1.424)		
Modified LIHC			-0.045	
			(0.057)	
Energy $\times$ Modified LIHC			-0.147	
			(0.119)	
NO2 Concentration				-0.054
				(0.054)
CO2 $\times$ NO2 Concentration				0.014
				(0.110)
PM210 Concentration				-0.123
				(0.093)
CO2 $\times$ PM210 Concentration				-0.130
				(0.188)
PM2.5 Concentration				0.089
				(0.100)

CO2 × PM2.5 Concentration				0.203 (0.204)
O3 Concentration				-0.026 (0.034)
CO2 × O3 Concentration				0.038 (0.068)
Constant	-6.353*** (1.364)	-3.107*** (0.355)	-2.993*** (0.464)	0.723 (2.814)
Observations	5255	5217	5255	5028

Note: Tobit regression estimation with standard errors clustered at the individual level. Dependent variable: *WTP*, ranging between $-\text{€}12.5$ and $\text{€}12.5$, computed as the midpoint of the interval in which the respondent shifts from one option to the other in the multiple price list, normalized as the difference from the baseline value (i.e., minus $\text{€}10$). Positive values indicate higher WTP (greater willingness to pay for first-hand). Each column estimates a separate specification including only the treatment relevant to the environmental context variable. Column (1): *Water* treatment interacted with provincial *Green vote share* (2024 EU elections). Column (2): *Water* treatment interacted with provincial *HPH* (hours of partial water suspension) and *Days of partial suspension* (dummy). Column (3): *Energy* treatment interacted with regional *Modified LIHC* (energy poverty index). Column (4): *CO2* treatment interacted with provincial air quality indicators (*NO₂*, *PM10*, *PM2.5*, *O₃* concentrations). Baseline: *Control* in all columns.

Significance of coefficients: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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